4615107

B.TECH. DEGREE EXAMINATION NOV/DEC 2014

Fifth semester – Mechanical Engineering

DESIGN OF MACHINE ELEMENTS

1. What is interchangeability?

Interchangeability occurs when one part in an assembly can be substituted for a similar part which has been made to the same drawing. Interchangeability is possible only when certain standards are strictly followed.

2. Explain hole basis and shaft basis system in tolerancing.

In this system, the basic diameter of the hole is constant while the shaft size is varied according to the type of fit.

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3. When the buttress thread is used?

It is used for transmission of power in one direction only. The force is transmitted almost parallel to the axis.

4. What is meant by fluctuating load?

The type of loads that vary continuously on the structure or a body are called fluctuating load. For example traffic passing on a bridge applies

5. What is the function of coupling? Name any two types of coupling.

- To provide for the connection of shafts of units that is manufactured separately and to provide for disconnection for repairs or alternations.
- To provide misalignment of the shafts or to introduce mechanical flexibility.
- To introduce protection against overloads.

Types of coupling,

- Sleeve or muff coupling
- Clamp or split muff or compression coupling.
- Flange coupling.

6. Why is it necessary to dissipate the heat generated when clutches operate

When a clutch gets hot the surfaces become more slippery and can cause them to warpand get heat spots which will make it jump and slip prematurely.

7. The extension spring is less used than compression springs why?

Perhaps because its easier to overextend the extension spring. Compression springs will “bottom out” before the over extend. Also it seems like the tensile strength wil be weaker at the attachment point for the extension spring, making it generally larger and more cumbersome to correct the deficiency.

8. What is nipping of laminated leaf spring? Discuss its roll in spring design.

Pre stressing of leaf springs is obtained by a difference of radii of curvature known as nipping.

The initial gap can be adjusted so that under max. load conditions the stress in all the leaves will be same or, if desired the stress is the full length leaves may be less.

9. What is the basis of shaft design when it is subjected to twisting moment only?

When the shaft is subjected to a twisting moment (or torque) only, then the diameter of the shaft may be obtained by using the torsion equation. We know that

$$T/J = \tau/r$$

10. What do you understand by torsional rigidity and lateral rigidity?

In the simplest of terms it is the resistance to twist. The modulus of rigidity (shear modulus) is the unit of measure of the force required to deform an object along its axis

Rigidity is a property by which an element resists deflection under the action of external forces or moments

Part B

11. The load on a bolt consists of an axial pull of 10 kN together with a transverse shear force of 5 kN. Find the diameter of bolt required according to 1. Maximum principal stress theory; 2. Maximum shear stress theory; 3. Maximum distortion energy theory.

Take permissible tensile stress at elastic limit = 100 MPa and poisson's ratio = 0.3.

Solution. Given : $P_{t1} = 10$ kN ; $P_s = 5$ kN ; $\sigma_{t(el)} = 100$ MPa $= 100$ N/mm^2 ; $1/m = 0.3$

Let d = Diameter of the bolt in mm.

∴ Cross-sectional area of the bolt,

$$A = \frac{\pi}{4} \times d^2 = 0.7854\, d^2 \text{ mm}^2$$

We know that axial tensile stress,

$$\sigma_1 = \frac{P_{t1}}{A} = \frac{10}{0.7854\, d^2} = \frac{12.73}{d^2} \text{ kN/mm}^2$$

and transverse shear stress,

$$\tau = \frac{P_s}{A} = \frac{5}{0.7854\, d^2} = \frac{6.365}{d^2} \text{ kN/mm}^2$$

1. According to maximum principal stress theory

We know that maximum principal stress,

$$\sigma_{t1} = \frac{\sigma_1 + \sigma_2}{2} + \frac{1}{2}\left[\sqrt{(\sigma_1 - \sigma_2)^2 + 4\tau^2}\right]$$

$$= \frac{\sigma_1}{2} + \frac{1}{2}\left[\sqrt{(\sigma_1)^2 + 4\tau^2}\right] \qquad \text{...}(\because \sigma_2 = 0)$$

$$= \frac{12.73}{2\, d^2} + \frac{1}{2}\left[\sqrt{\left(\frac{12.73}{d^2}\right)^2 + 4\left(\frac{6.365}{d^2}\right)^2}\right]$$

$$= \frac{6.365}{d^2} + \frac{1}{2} \times \frac{6.365}{d^2}\left[\sqrt{4+4}\right]$$

$$= \frac{6.365}{d^2}\left[1 + \frac{1}{2}\sqrt{4+4}\right] = \frac{15.365}{d^2} \text{ kN/mm}^2 = \frac{15\,365}{d^2} \text{ N/mm}^2$$

According to maximum principal stress theory,

$$\sigma_{t1} = \sigma_{t(el)} \quad \text{or} \quad \frac{15\,365}{d^2} = 100$$

∴ $d^2 = 15\,365/100 = 153.65$ or $d = 12.4$ mm **Ans.**

2. *According to maximum shear stress theory*

We know that maximum shear stress,

$$\tau_{max} = \frac{1}{2}\left[\sqrt{(\sigma_1 - \sigma_2)^2 + 4\,\tau^2}\right] = \frac{1}{2}\left[\sqrt{(\sigma_1)^2 + 4\,\tau^2}\right] \qquad ...(\because \sigma_2 = 0)$$

$$= \frac{1}{2}\left[\sqrt{\left(\frac{12.73}{d^2}\right)^2 + 4\left(\frac{6.365}{d^2}\right)^2}\right] = \frac{1}{2} \times \frac{6.365}{d^2}\left[\sqrt{4+4}\right]$$

$$= \frac{9}{d^2}\ \text{kN/mm}^2 = \frac{9000}{d^2}\ \text{N/mm}^2$$

According to maximum shear stress theory,

$$\tau_{max} = \frac{\sigma_{t(el)}}{2} \quad \text{or} \quad \frac{9000}{d^2} = \frac{100}{2} = 50$$

$\therefore$ $d^2 = 9000 / 50 = 180$ or $d = 13.42$ mm **Ans.**

5. *According to maximum distortion energy theory*

According to maximum distortion energy theory,

$$(\sigma_{t1})^2 + (\sigma_{t2})^2 - 2\sigma_{t1} \times \sigma_{t2} = [\sigma_{t(el)}]^2$$

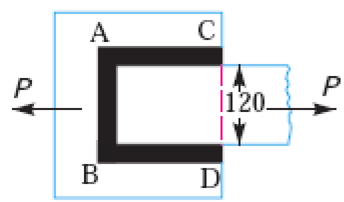
$$\left[\frac{15\,365}{d^2}\right]^2 + \left[\frac{-2635}{d^2}\right]^2 - 2 \times \frac{15\,365}{d^2} \times \frac{-2635}{d^2} = (100)^2$$

$$\frac{236 \times 10^6}{d^4} + \frac{6.94 \times 10^6}{d^4} + \frac{80.97 \times 10^6}{d^4} = 10 \times 10^3$$

$$\frac{23\,600}{d^4} + \frac{694}{d^4} + \frac{8097}{d^4} = 1 \quad \text{or} \quad \frac{32\,391}{d^4} = 1$$

$\therefore$ $d^4 = 32\,391$ or $d = 13.4$ mm **Ans.**

12. Determine the length of the weld run for a plate of size 120 mm wide and 15 mm thick to be welded to another plate by means of 1. A single transverse weld; and 2. Double parallel fillet welds when the joint is subjected to variable loads.



Given : Width = 120 mm ; Thickness = 15 mm In Fig. *AB* represents the single transverse weld and *AC* and *BD* represents double parallel fillet welds.

1. *Length of the weld run for a single transverse weld*

The effective length of the weld run (l_1) for a single transverse weld may be obtained by subtracting 12.5 mm from the width of the plate.

$\therefore$ $l_1 = 120 - 12.5 = 107.5$ mm **Ans.**

2. *Length of the weld run for a double parallel fillet weld subjected to variable loads*

Let l_2 = Length of weld run for each parallel fillet, and

s = Size of weld = Thickness of plate = 15 mm

Assuming the tensile stress as 70 MPa or N/mm^2 and shear stress as 56 MPa or N/mm^2 for static loading. We know that the maximum load which the plate can carry is

$$P = \text{Area} \times \text{Stress} = 120 \times 15 \times 70 = 126 \times 10^3 \text{ N}$$

From Table 10.6, we find that the stress concentration factor for transverse weld is 1.5 and for parallel fillet welds is 2.7.

∴ Permissible tensile stress,

$$\sigma_t = 70 / 1.5 = 46.7 \text{ N/mm}^2$$

and permissible shear stress,

$$\tau = 56 / 2.7 = 20.74 \text{ N/mm}^2$$

∴ Load carried by single transverse weld,

$$P_1 = 0.707\, s \times l_1 \times \sigma_t = 0.707 \times 15 \times 107.5 \times 46.7 = 53\,240 \text{ N}$$

and load carried by double parallel fillet weld,

$$P_2 = 1.414\, s \times l_2 \times \tau = 1.414 \times 15 \times l_2 \times 20.74 = 440\, l_2 \text{ N}$$

∴ Load carried by the joint (P),

$$126 \times 10^3 = P_1 + P_2 = 53\,240 + 440\, l_2 \quad \text{or} \quad l_2 = 165.4 \text{ mm}$$

Adding 12.5 mm for starting and stopping of weld run, we have

$$l_2 = 165.4 + 12.5 = 177.9 \text{ say } 178 \text{ mm}$$ **Ans.**

13. An electric motor driven power screw moves a nut in a horizontal plane against a force of 75 kN at a speed of 300 mm / min. The screw has a single square thread of 6 mm pitch on a major diameter of 40 mm. The coefficient of friction at screw threads is 0.1. Estimate power of the motor.

Solution. Given : $W = 75$ kN $= 75 \times 10^3$ N ; $v = 300$ mm/min ; $p = 6$ mm ; $d_o = 40$ mm ; $\mu = \tan\phi = 0.1$

We know that mean diameter of the screw,

$$d = d_o - p/2 = 40 - 6/2 = 37 \text{ mm}$$

and

$$\tan\alpha = \frac{p}{\pi d} = \frac{6}{\pi \times 37} = 0.0516$$

We know that tangential force required at the circumference of the screw,

$$P = W\tan(\alpha + \phi) = W\left[\frac{\tan\alpha + \tan\phi}{1 - \tan\alpha\tan\phi}\right]$$

$$= 75 \times 10^3 \left[\frac{0.0516 + 0.1}{1 - 0.0516 \times 0.1}\right] = 11.43 \times 10^3 \text{ N}$$

and torque required to operate the screw,

$$T = P \times \frac{d}{2} = 11.43 \times 10^3 \times \frac{37}{2} = 211.45 \times 10^3 \text{ N-mm} = 211.45 \text{ N-m}$$

Since the screw moves in a nut at a speed of 300 mm / min and the pitch of the screw is 6 mm therefore speed of the screw in revolutions per minute (r.p.m.),

$$N = \frac{\text{Speed in mm/min.}}{\text{Pitch in mm}} = \frac{300}{6} = 50 \text{ r.p.m.}$$

and angular speed, $\omega = 2\pi N / 60 = 2\pi \times 50 / 60 = 5.24$ rad /s

∴ Power of the motor $= T.\omega = 211.45 \times 5.24 = 1108$ W $= 1.108$ kW **Ans.**

14. A bracket, as shown in Fig. above, supports a load of 30 kN. Determine the size of bolts, if the maximum allowable tensile stress in the bolt material is 60 MPa. The distances are: L_1 = 80 mm, L_2 = 250 mm, and L = 500 mm.

Solution. Given : $W = 30$ kN ; $\sigma_t = 60$ MPa = 60 N/mm^2 ; $L_1 = 80$ mm ; $L_2 = 250$ mm ; $L = 500$ mm

We know that the direct tensile load carried by each bolt,

$$W_{t1} = \frac{W}{n} = \frac{30}{4} = 7.5 \text{ kN}$$

and load in a bolt per unit distance,

$$w = \frac{W.L}{2\,[(L_1)^2 + (L_2)^2]} = \frac{30 \times 500}{2\,[(80)^2 + (250)^2]} = 0.109 \text{ kN/mm}$$

Since the heavily loaded bolt is at a distance of L_2 mm from the tilting edge, therefore load on the heavily loaded bolt,

$$W_{t2} = w.L_2 = 0.109 \times 250 = 27.25 \text{ kN}$$

∴ Maximum tensile load on the heavily loaded bolt,

$$W_t = W_{t1} + W_{t2} = 7.5 + 27.25 = 34.75 \text{ kN} = 34\,750 \text{ N}$$

Let d_c = Core diameter of the bolts.

We know that the maximum tensile load on the bolt (W_t),

$$34\,750 = \frac{\pi}{4}(d_c)^2\,\sigma_t = \frac{\pi}{4}(d_c)^2\,60 = 47\,(d_c)^2$$

∴ $(d_c)^2 = 34\,750 / 47 = 740$

From Table (coarse series), we find that the standard core diameter of the bolt is 28.706 mm and the corresponding size of the bolt is M 33.

15. Design a cast iron protective type flange coupling to transmit 15 kW at 900r.p.m. from an electric motor to a compressor. The service factor may be assumed as 1.35. The following permissible stresses may be used: Shear stress for shaft, bolt and key material = 40 Mpa, Crushing stress for bolt and key = 80 Mpa, Shear stress for cast iron = 8 Mpa, Draw a neat sketch of the coupling.

Solution. Given : $P = 15$ kW $= 15 \times 10^3$ W ; $N = 900$ r.p.m. ; Service factor = 1.35 ; $\tau_s = \tau_b = \tau_k = 40$ MPa = 40 N/mm^2 ; $\sigma_{cb} = \sigma_{ck} = 80$ MPa = 80 N/mm^2 ; $\tau_c = 8$ MPa = 8 N/mm^2

The protective type flange coupling is designed as discussed below :

1. Design for hub

First of all, let us find the diameter of the shaft (d). We know that the torque transmitted by the shaft,

$$T = \frac{P \times 60}{2\,\pi N} = \frac{15 \times 10^3 \times 60}{2\,\pi \times 900} = 159.13 \text{ N-m}$$

Since the service factor is 1.35, therefore the maximum torque transmitted by the shaft,

$$T_{max} = 1.35 \times 159.13 = 215 \text{ N-m} = 215 \times 10^3 \text{ N-mm}$$

We know that the torque transmitted by the shaft (T),

$$215 \times 10^3 = \frac{\pi}{16} \times \tau_s \times d^3 = \frac{\pi}{16} \times 40 \times d^3 = 7.86\, d^3$$

$\therefore$ $d^3 = 215 \times 10^3 / 7.86 = 27.4 \times 10^3$ or $d = 30.1$ say 35 mm **Ans.**

We know that outer diameter of the hub,

$$D = 2d = 2 \times 35 = 70 \text{ mm } \textbf{Ans.}$$

and length of hub, $L = 1.5\, d = 1.5 \times 35 = 52.5$ mm **Ans.**

Let us now check the induced shear stress for the hub material which is cast iron. Considering the hub as a hollow shaft. We know that the maximum torque transmitted (T_{max}).

$$215 \times 10^3 = \frac{\pi}{16} \times \tau_c \left[\frac{D^4 - d^4}{D}\right] = \frac{\pi}{16} \times \tau_c \left[\frac{(70)^4 - (35)^4}{70}\right] = 63\ 147\ \tau_c$$

$\therefore$ $\tau_c = 215 \times 10^3 / 63\ 147 = 3.4 \text{ N/mm}^2 = 3.4$ MPa

Since the induced shear stress for the hub material (*i.e.* cast iron) is less than the permissible value of 8 MPa, therefore the design of hub is safe.

2. *Design for key*

Since the crushing stress for the key material is twice its shear stress (*i.e.* $\sigma_{ck} = 2\tau_k$), therefore a square key may be used. From Table 13.1, we find that for a shaft of 35 mm diameter,

Width of key, $w = 12$ mm **Ans.**

and thickness of key, $t = w = 12$ mm **Ans.**

The length of key (l) is taken equal to the length of hub.

$\therefore$ $l = L = 52.5$ mm **Ans.**

Let us now check the induced stresses in the key by considering it in shearing and crushing.

Considering the key in shearing. We know that the maximum torque transmitted (T_{max}),

$$215 \times 10^3 = l \times w \times \tau_k \times \frac{d}{2} = 52.5 \times 12 \times \tau_k \times \frac{35}{2} = 11\ 025\ \tau_k$$

$\therefore$ $\tau_k = 215 \times 10^3 / 11\ 025 = 19.5 \text{ N/mm}^2 = 19.5$ MPa

Considering the key in crushing. We know that the maximum torque transmitted (T_{max}),

$$215 \times 10^3 = l \times \frac{t}{2} \times \sigma_{ck} \times \frac{d}{2} = 52.5 \times \frac{12}{2} \times \sigma_{ck} \times \frac{35}{2} = 5512.5\ \sigma_{ck}$$

$\therefore$ $\sigma_{ck} = 215 \times 10^3 / 5512.5 = 39 \text{ N/mm}^2 = 39$ MPa

Since the induced shear and crushing stresses in the key are less than the permissible stresses, therefore the design for key is safe.

3. *Design for flange*

The thickness of flange (t_f) is taken as 0.5 d.

$\therefore$ $t_f = 0.5\, d = 0.5 \times 35 = 17.5$ mm **Ans.**

Let us now check the induced shearing stress in the flange by considering the flange at the junction of the hub in shear.

We know that the maximum torque transmitted (T_{max}),

$$215 \times 10^3 = \frac{\pi D^2}{2} \times \tau_c \times t_f = \frac{\pi (70)^2}{2} \times \tau_c \times 17.5 = 134\ 713\ \tau_c$$

$\therefore$ $\tau_c = 215 \times 10^3/134\ 713 = 1.6\ \text{N/mm}^2 = 1.6\ \text{MPa}$

Since the induced shear stress in the flange is less than 8 MPa, therefore the design of flange is safe.

4. *Design for bolts*

Let d_1 = Nominal diameter of bolts.

Since the diameter of the shaft is 35 mm, therefore let us take the number of bolts,

$$n = 3$$

and pitch circle diameter of bolts,

$$D_1 = 3d = 3 \times 35 = 105\ \text{mm}$$

The bolts are subjected to shear stress due to the torque transmitted. We know that the maximum torque transmitted (T_{max}),

$$215 \times 10^3 = \frac{\pi}{4} (d_1)^2\ \tau_b \times n \times \frac{D_1}{2} = \frac{\pi}{4} (d_1)^2\ 40 \times 3 \times \frac{105}{2} = 4950\ (d_1)^2$$

$\therefore$ $(d_1)^2 = 215 \times 10^3/4950 = 43.43$ or $d_1 = 6.6$ mm

Assuming coarse threads, the nearest standard size of bolt is M 8. **Ans.**

Other proportions of the flange are taken as follows :

Outer diameter of the flange,

$$D_2 = 4\ d = 4 \times 35 = 140\ \text{mm} \quad \textbf{Ans.}$$

Thickness of the protective circumferential flange,

$$t_p = 0.25\ d = 0.25 \times 35 = 8.75 \text{ say } 10\ \text{mm} \quad \textbf{Ans.}$$

16. A single plate clutch, effective on both sides, is required to transmit 25 kW at 3000 r.p.m. Determine the outer and inner diameters of frictional surface if the coefficient of friction is 0.255, ratio of diameters is 1.25 and the maximum pressure is not to exceed 0.1 N/mm². Also, determine the axial thrust to be provided by springs. Assume the theory of uniform wear.

Solution. Given : $n = 2$; $P = 25$ kW $= 25 \times 10^3$ W ; $N = 3000$ r.p.m. ; $\mu = 0.255$; $d_1 / d_2 = 1.25$ or $r_1 / r_2 = 1.25$; $p_{max} = 0.1\ \text{N/mm}^2$

Outer and inner diameters of frictional surface

Let d_1 and d_2 = Outer and inner diameters (in mm) of frictional surface, and

r_1 and r_2 = Corresponding radii (in mm) of frictional surface.

We know that the torque transmitted by the clutch,

$$T = \frac{P \times 60}{2\ \pi\ N} = \frac{25 \times 10^3 \times 60}{2\ \pi \times 3000} = 79.6\ \text{N-m} = 79\ 600\ \text{N-mm}$$

For uniform wear conditions, $p.r = C$ (a constant). Since the intensity of pressure is maximum at the inner radius (r_2), therefore

$$p_{max} \times r_2 = C$$

or $C = 0.1\ r_2$ N/mm

and normal or axial load acting on the friction surface,

$$W = 2\pi\, C\,(r_1 - r_2) = 2\pi \times 0.1\, r_2\,(1.25\, r_2 - r_2)$$
$$= 0.157\,(r_2)^2 \qquad \ldots (\because\ r_1 / r_2 = 1.25)$$

We know that mean radius of the frictional surface (for uniform wear),

$$R = \frac{r_1 + r_2}{2} = \frac{1.25\, r_2 + r_2}{2} = 1.125\, r_2$$

and the torque transmitted (T),

$$79\,600 = n.\mu.W.R = 2 \times 0.255 \times 0.157\,(r_2)^2\, 1.125\, r_2 = 0.09\,(r_2)^3$$

$$\therefore \quad (r_2)^3 = 79.6 \times 10^3 / 0.09 = 884 \times 10^3 \text{ or } r_2 = 96 \text{ mm}$$

and $\quad r_1 = 1.25\, r_2 = 1.25 \times 96 = 120$ mm

$\therefore$ Outer diameter of frictional surface,

$$d_1 = 2r_1 = 2 \times 120 = 240 \text{ mm} \quad \textbf{Ans.}$$

and inner diameter of frictional surface,

$$d_2 = 2r_2 = 2 \times 96 = 192 \text{ mm} \quad \textbf{Ans.}$$

Axial thrust to be provided by springs

We know that axial thrust to be provided by springs,

$$W = 2\pi\, C\,(r_1 - r_2) = 2\pi \times 0.1\, r_2\,(1.25\, r_2 - r_2)$$
$$= 0.157\,(r_2)^2 = 0.157\,(96)^2 = 1447 \text{ N} \quad \textbf{Ans.}$$

17. A rail wagon of mass 20 tones is moving with a velocity of 2 m/s. It is brought to rest by two buffers with springs of 300 mm diameter. The maximum deflection of springs is 250 mm. The allowable shear stress in the spring material is 600 MPa. Design the spring for the buffers.

Given: m = 20 t = 20 000 kg; v = 2 m/s; D = 300 mm; δ = 250 mm; τ= 600 MPa = 600 N/mm^2

1. Diameter of the spring wire

Let d =Diameter of the spring wire.

We know that kinetic energy of the wagon

$$= \frac{1}{2}\, m\, v^2 = \frac{1}{2} \times 20\,000\,(2)^2 = 40\,000 \text{ N-m} = 40 \times 10^6 \text{ N-mm} \qquad \ldots(i)$$

Let W be the equivalent load which when applied gradually on each spring causes a deflection of 250 mm. Since there are two springs, therefore

Energy stored in the springs

$$= \frac{1}{2} \times W.\delta \times 2 = W.\delta = W \times 250 = 250\, W \text{ N-mm} \qquad \ldots(ii)$$

From equations (*i*) and (*ii*), we have

$$\therefore \quad W = 40 \times 10^6 / 250 = 160 \times 10^3 \text{ N}$$

We know that torque transmitted by the spring,

$$T = W \times \frac{D}{2} = 160 \times 10^3 \times \frac{300}{2} = 24 \times 10^6 \text{ N-mm}$$

We also know that torque transmitted by the spring (T),

$$24 \times 10^6 = \frac{\pi}{16} \times \tau \times d^3 = \frac{\pi}{16} \times 600 \times d^3 = 117.8\ d^3$$

$$\therefore \quad d^3 = 24 \times 10^6 / 117.8 = 203.7 \times 10^3 \quad \text{or} \quad d = 58.8 \text{ say } 60 \text{ mm } \textbf{Ans.}$$

2. Number of turns of the spring coil

Let n = Number of active turns of the spring coil.

We know that the deflection of the spring (δ),

$$250 = \frac{8\ W.D^3.n}{G.d^4} = \frac{8 \times 160 \times 10^3\ (300)^3\ n}{84 \times 10^3\ (60)^4} = 31.7\ n$$

... (Taking $G = 84$ MPa $= 84 \times 10^3$ N/mm^2)

$$\therefore \quad n = 250 / 31.7 = 7.88 \text{ say } 8 \textbf{ Ans.}$$

Assuming square and ground ends, total number of turns,

$$n' = n + 2 = 8 + 2 = 10 \textbf{ Ans.}$$

3. Free length of the spring

We know that free length of the spring,

$$L_F = n'.d + \delta + 0.15\ \delta = 10 \times 60 + 250 + 0.15 \times 250 = 887.5 \text{ mm } \textbf{Ans.}$$

4. Pitch of the coil

We know that pitch of the coil

$$= \frac{\text{Free length}}{n' - 1} = \frac{887.5}{10 - 1} = 98.6 \text{ mm } \textbf{Ans.}$$

18. Design and draw a cotter joint to support a load varying from 30 kN in compression to 30 kN in tension. The material used is carbon steel for which the following allowable stresses may be used. The load is applied statically. Tensile stress = compressive stress = 50 MPa; shear stress = 35 MPa and crushing stress = 90 MPa.

Solution. Given: P = 30 kN = 30×10^3 N; σ_t = 50 MPa = 50 N / mm^2; τ = 35 MPa = 35 N/ mm^2; σ_c = 90 MPa = 90 N/mm^2

1. Diameter of the rods

Let d = Diameter of the rods.

Considering the failure of the rod in tension. We know that load (P),

$$30 \times 10^3 = \frac{\pi}{4} \times d^2 \times \sigma_t = \frac{\pi}{4} \times d^2 \times 50 = 39.3\ d^2$$

$$\therefore \quad d^2 = 30 \times 10^3 / 39.3 = 763 \quad \text{or} \quad d = 27.6 \text{ say } 28 \text{ mm } \textbf{Ans.}$$

2. Diameter of spigot and thickness of cotter

Let d2 = Diameter of spigot or inside diameter of socket, and

t = Thickness of cotter. It may be taken as d2 / 4.

Let $\qquad d_2$ = Diameter of spigot or inside diameter of socket, and

$\qquad t$ = Thickness of cotter. It may be taken as $d_2 / 4$.

Considering the failure of spigot in tension across the weakest section. We know that load (P),

$$30 \times 10^3 = \left[\frac{\pi}{4}(d_2)^2 - d_2 \times t\right]\sigma_t = \left[\frac{\pi}{4}(d_2)^2 - d_2 \times \frac{d_2}{4}\right]50 = 26.8\,(d_2)^2$$

$$\therefore \qquad (d_2)^2 = 30 \times 10^3 / 26.8 = 1119.4 \quad \text{or} \quad d_2 = 33.4 \text{ say } 34 \text{ mm}$$

and thickness of cotter, $t = \dfrac{d_2}{4} = \dfrac{34}{4} = 8.5$ mm

Let us now check the induced crushing stress. We know that load (P),

$$30 \times 10^3 = d_2 \times t \times \sigma_c = 34 \times 8.5 \times \sigma_c = 289\,\sigma_c$$

$$\therefore \qquad \sigma_c = 30 \times 10^3 / 289 = 103.8 \text{ N/mm}^2$$

Since this value of σ_c is more than the given value of $\sigma_c = 90$ N/mm^2, therefore the dimensions d_2 = 34 mm and t = 8.5 mm are not safe. Now let us find the values of d_2 and t by substituting the value of $\sigma_c = 90$ N/mm^2 in the above expression, *i.e.*

$$30 \times 10^3 = d_2 \times \frac{d_2}{4} \times 90 = 22.5\,(d_2)^2$$

$$\therefore \qquad (d_2)^2 = 30 \times 10^3 / 22.5 = 1333 \quad \text{or} \quad d_2 = 36.5 \text{ say } 40 \text{ mm}$$ **Ans.**

and $\qquad t = d_2 / 4 = 40 / 4 = 10$ mm **Ans.**

3. Outside diameter of socket

Let $\qquad d_1$ = Outside diameter of socket.

Considering the failure of the socket in tension across the slot. We know that load (P),

$$30 \times 10^3 = \left[\frac{\pi}{4}\{(d_1)^2 - (d_2)^2\} - (d_1 - d_2)\,t\right]\sigma_t$$

$$= \left[\frac{\pi}{4}\{(d_1)^2 - (40)^2\} - (d_1 - 40)\,10\right]50$$

$$30 \times 10^3/50 = 0.7854\,(d_1)^2 - 1256.6 - 10\,d_1 + 400$$

$$\text{or} \quad (d_1)^2 - 12.7\,d_1 - 1854.6 = 0$$

$$\therefore \qquad d_1 = \frac{12.7 \pm \sqrt{(12.7)^2 + 4 \times 1854.6}}{2} = \frac{12.7 \pm 87.1}{2}$$

$$= 49.9 \text{ say } 50 \text{ mm}$$ **Ans.** ...(Taking +ve sign)

4. Width of cotter

Let $\qquad b$ = Width of cotter.

Considering the failure of the cotter in shear. Since the cotter is in double shear, therefore load (P),

$$30 \times 10^3 = 2\,b \times t \times \tau = 2\,b \times 10 \times 35 = 700\,b$$

$\therefore \quad b = 30 \times 10^3 / 700 = 43$ mm **Ans.**

5. Diameter of socket collar

Let $\quad d_4$ = Diameter of socket collar.

Considering the failure of the socket collar and cotter in crushing. We know that load (P),

$$30 \times 10^3 = (d_4 - d_2)\,t \times \sigma_c = (d_4 - 40)10 \times 90 = (d_4 - 40)\,900$$

$\therefore \quad d_4 - 40 = 30 \times 10^3 / 900 = 33.3$ or $d_4 = 33.3 + 40 = 73.3$ say 75 mm **Ans.**

6. Thickness of socket collar

Let $\quad c$ = Thickness of socket collar.

Considering the failure of the socket end in shearing. Since the socket end is in double shear, therefore load (P),

$$30 \times 10^3 = 2(d_4 - d_2)\,c \times \tau = 2\,(75 - 40)\,c \times 35 = 2450\,c$$

$\therefore \quad c = 30 \times 10^3 / 2450 = 12$ mm **Ans.**

7. Distance from the end of the slot to the end of the rod

Let $\quad a$ = Distance from the end of slot to the end of the rod.

Considering the failure of the rod end in shear. Since the rod end is in double shear, therefore load (P),

$$30 \times 10^3 = 2\,a \times d_2 \times \tau = 2a \times 40 \times 35 = 2800\,a$$

$\therefore \quad a = 30 \times 10^3 / 2800 = 10.7$ say 11 mm **Ans.**

8. Diameter of spigot collar

Let $\quad d_3$ = Diameter of spigot collar.

Considering the failure of spigot collar in crushing. We know that load (P),

$$30 \times 10^3 = \frac{\pi}{4}\left[(d_3)^2 - (d_2)^2\right]\sigma_c = \frac{\pi}{4}\left[(d_3)^2 - (40)^2\right]90$$

or $$(d_3)^2 - (40)^2 = \frac{30 \times 10^3 \times 4}{90 \times \pi} = 424$$

$\therefore \quad (d_3)^2 = 424 + (40)^2 = 2024$ or $d_3 = 45$ mm **Ans.**

9. Thickness of spigot collar

Let $\quad t_1$ = Thickness of spigot collar.

Considering the failure of spigot collar in shearing. We know that load (P)

$$30 \times 10^3 = \pi\,d_2 \times t_1 \times \tau = \pi \times 40 \times t_1 \times 35 = 4400\,t_1$$

$\therefore \quad t_1 = 30 \times 10^3 / 4400 = 6.8$ say 8 mm **Ans.**

10. The length of cotter (l) is taken as 4 d.

$\therefore \quad l = 4\,d = 4 \times 28 = 112$ mm **Ans.**

11. The dimension e is taken as 1.2 d.

$\therefore \quad e = 1.2 \times 28 = 33.6$ say 34 mm **Ans.**

19. A machine component is subjected to a flexural stress which fluctuates between + 300 MN/m2 and – 150 MN/m2. Determine the value of minimum ultimate strength according to

Modified Goodman relation; and Soderberg relation. Take yield strength = 0.55 Ultimate strength; Endurance strength = 0.5 Ultimate strength; and factor of safety = 2.

Solution. Given : σ_1 = 300 MN/m^2 ; $\sigma_2 = -150$ MN/m^2 ; $\sigma_y = 0.55\ \sigma_u$; $\sigma_e = 0.5\ \sigma_u$; $F.S. = 2$

Let σ_u = Minimum ultimate strength in MN/m^2.

We know that the mean or average stress,

$$\sigma_m = \frac{\sigma_1 + \sigma_2}{2} = \frac{300 + (-150)}{2} = 75 \text{ MN/m}^2$$

and variable stress,

$$\sigma_v = \frac{\sigma_1 - \sigma_2}{2} = \frac{300 - (-150)}{2} = 225 \text{ MN/m}^2$$

According to modified Goodman relation

We know that according to modified Goodman relation,

$$\frac{1}{F.S.} = \frac{\sigma_m}{\sigma_u} + \frac{\sigma_v}{\sigma_e}$$

or

$$\frac{1}{2} = \frac{75}{\sigma_u} + \frac{225}{0.5\ \sigma_u} = \frac{525}{\sigma_u}$$

$\therefore$ $\sigma_u = 2 \times 525 = 1050$ MN/m^2 **Ans.**

According to Soderberg relation

We know that according to Soderberg relation,

$$\frac{1}{F.S.} = \frac{\sigma_m}{\sigma_y} + \frac{\sigma_v}{\sigma_e}$$

or

$$\frac{1}{2} = \frac{75}{0.55\ \sigma_u} + \frac{255}{0.5\ \sigma_u} = \frac{586.36}{\sigma_u}$$

$\therefore$ $\sigma_u = 2 \times 586.36 = 1172.72$ MN/m^2 **Ans.**

20. A solid circular shaft is subjected to a bending moment of 3000 N-m and a torque of 10 000 N-m. The shaft is made of 45 C 8 steel having ultimate tensile stress of 700 MPa and a ultimate shear stress of 500 MPa. Assuming a factor of safety as 6, determine the diameter of the shaft.

Solution. Given : M = 3000 N-m = 3 × 10^6 N-mm ; T = 10 000 N-m = 10 × 10^6 N-mm ; σ_{tu} = 700 MPa = 700 N/mm^2 ; τ_u = 500 MPa = 500 N/mm^2

We know that the allowable tensile stress,

$$\sigma_t \text{ or } \sigma_b = \frac{\sigma_{tu}}{F.S.} = \frac{700}{6} = 116.7 \text{ N/mm}^2$$

and allowable shear stress,

$$\tau = \frac{\tau_u}{F.S.} = \frac{500}{6} = 83.3 \text{ N/mm}^2$$

Let d = Diameter of the shaft in mm.

According to maximum shear stress theory, equivalent twisting moment,

$$T_e = \sqrt{M^2 + T^2} = \sqrt{(3 \times 10^6)^2 + (10 \times 10^6)^2} = 10.44 \times 10^6 \text{ N-mm}$$

We also know that equivalent twisting moment (T_e),

$$10.44 \times 10^6 = \frac{\pi}{16} \times \tau \times d^3 = \frac{\pi}{16} \times 83.3 \times d^3 = 16.36\ d^3$$

$$\therefore \quad d^3 = 10.44 \times 10^6 / 16.36 = 0.636 \times 10^6 \text{ or } d = 86 \text{ mm}$$

According to maximum normal stress theory, equivalent bending moment,

$$M_e = \tfrac{1}{2}\left(M + \sqrt{M^2 + T^2}\right) = \tfrac{1}{2}(M + T_e)$$

$$= \tfrac{1}{2}(3 \times 10^6 + 10.44 \times 10^6) = 6.72 \times 10^6 \text{ N-mm}$$

We also know that the equivalent bending moment (M_e),

$$6.72 \times 10^6 = \frac{\pi}{32} \times \sigma_b \times d^3 = \frac{\pi}{32} \times 116.7 \times d^3 = 11.46\ d^3$$

$$\therefore \quad d^3 = 6.72 \times 10^6 / 11.46 = 0.586 \times 10^6 \text{ or } d = 83.7 \text{ mm}$$

Taking the larger of the two values, we have

$d = 86$ say 90 mm **Ans.**

<u>**5615121**</u>

B.TECH. DEGREE EXAMINATION NOV/DEC 2015

Fifth semester – Mechanical Engineering

DESIGN OF MACHINE ELEMENTS

<u>**Part A**</u>

1. **Define factor of safety.**

Factor of safety (FOS) is defined as the ratio between the Ultimate stress and working stress.

2. **What are preferred numbers?**

The preferred numbers are the conventionally rounded off values derived from geometric series including the integral powers of 10 and having a common ratio of the following factors. Basic series are R5, R10, R20, R40 & R80

3. **What is the meant by leak proof joint?**

The bolts selected for joint must sufficient contact pressure on the gasket to hold the leakage in the joint without causing any overloading of the bolt and the flange.

4. **State the various types of screw threads used for power screws.**

Following are the three types of screw threads mostly used for power screws:

- ✓ Square thread
- ✓ Acme or trapezoidal thread
- ✓ Buttress thread

5. **State some of the requirements of a shaft coupling?**

- ✓ It should be easy to connect or disconnect.
- ✓ It should transmit the full power of the shaft
- ✓ It should hold the shafts in perfect alignment.
- ✓ It should have no projecting parts.

6. **What is the purpose of a clutch?**

A clutch is a machine member used to connect a driving shaft to a driven shaft so that the driven shaft may be started or stopped at will, without stopping the driving shaft

7. **Define "Spring Index".**

It is the ratio of mean pitch diameter to the diameter of the wire(C= D/d).

8. **What are the various types of cotter joints?**

Following are the three commonly used cotter joints to connect two rods by a cotter:

- ✓ Socket and spigot cotter joint,
- ✓ Sleeve and cotter joint, and
- ✓ Gib and cotter joint.

9. **Define "Resilience".**

Resilience is the property of the material to absorb energy and to resist shock and impact loads. This property is essential for spring materials.

10. **Define "stress concentration factor".**

Stress concentration factor = maximum stress at the change of cross section / nominal stress.

Part B

11. A cylindrical shaft made of steel of yield strength 700 MPa is subjected to static loads consisting of bending moment 10 kN-m and a torsional moment 30 kN-m. Determine the diameter of the shaft using two different theories of failure, and assuming a factor of safety of 2. Take E = 210 GPa and poisson's ratio = 0.25

Solution. Given : $\sigma_{yt} = 700$ MPa $= 700$ N/mm^2 ; $M = 10$ kN-m $= 10 \times 10^6$ N-mm ; $T = 30$ kN-m $= 30 \times 10^6$ N-mm ; $F.S. = 2$; $E = 210$ GPa $= 210 \times 10^3$ N/mm^2 ; $1/m = 0.25$

Let d = Diameter of the shaft in mm.

First of all, let us find the maximum and minimum principal stresses.

We know that section modulus of the shaft

$$Z = \frac{\pi}{32} \times d^3 = 0.0982\, d^3 \text{ mm}^3$$

$\therefore$ Bending (tensile) stress due to the bending moment,

$$\sigma_1 = \frac{M}{Z} = \frac{10 \times 10^6}{0.0982\, d^3} = \frac{101.8 \times 10^6}{d^3} \text{ N/mm}^2$$

and shear stress due to torsional moment,

$$\tau = \frac{16\, T}{\pi\, d^3} = \frac{16 \times 30 \times 10^6}{\pi\, d^3} = \frac{152.8 \times 10^6}{d^3} \text{ N/mm}^2$$

We know that maximum principal stress,

$$\sigma_{t1} = \frac{\sigma_1 + \sigma_2}{2} + \frac{1}{2}\left[\sqrt{(\sigma_1 - \sigma_2)^2 + 4\,\tau^2}\right]$$

$$= \frac{\sigma_1}{2} + \frac{1}{2}\left[\sqrt{(\sigma_1)^2 + 4\,\tau^2}\right] \qquad ...(\because \sigma_2 = 0)$$

$$= \frac{101.8 \times 10^6}{2d^3} + \frac{1}{2}\left[\sqrt{\left(\frac{101.8 \times 10^6}{d^3}\right)^2 + 4\left(\frac{152.8 \times 10^6}{d^3}\right)^2}\right]$$

$$= \frac{50.9 \times 10^6}{d^3} + \frac{1}{2} \times \frac{10^6}{d^3}\left[\sqrt{(101.8)^2 + 4\,(152.8)^2}\right]$$

$$= \frac{50.9 \times 10^6}{d^3} + \frac{161 \times 10^6}{d^3} = \frac{211.9 \times 10^6}{d^3} \text{ N/mm}^2$$

and minimum principal stress,

$$\sigma_{t2} = \frac{\sigma_1 + \sigma_2}{2} - \frac{1}{2}\left[\sqrt{(\sigma_1 - \sigma_2)^2 + 4\,\tau^2}\right]$$

$$= \frac{\sigma_1}{2} - \frac{1}{2}\left[\sqrt{(\sigma_1)^2 + 4\,\tau^2}\right] \qquad ...(\because \sigma_2 = 0)$$

$$= \frac{50.9 \times 10^6}{d^3} - \frac{161 \times 10^6}{d^3} = \frac{-110.1 \times 10^6}{d^3} \text{ N/mm}^2$$

Let us now find out the diameter of shaft (d) by considering the maximum shear stress theory and maximum strain energy theory.

1. *According to maximum shear stress theory*

We know that maximum shear stress,

$$\tau_{max} = \frac{\sigma_{t1} - \sigma_{t2}}{2} = \frac{1}{2}\left[\frac{211.9 \times 10^6}{d^3} + \frac{110.1 \times 10^6}{d^3}\right] = \frac{161 \times 10^6}{d^3}$$

We also know that according to maximum shear stress theory,

$$\tau_{max} = \frac{\sigma_{yt}}{2\ F.S.} \quad \text{or} \quad \frac{161 \times 10^6}{d^3} = \frac{700}{2 \times 2} = 175$$

$$\therefore \quad d^3 = 161 \times 10^6 / 175 = 920 \times 10^3 \text{ or } d = 97.2 \text{ mm } \textbf{Ans.}$$

2. *According to maximum strain energy theory*

We know that according to maximum strain energy theory,

$$\frac{1}{2E}\left[(\sigma_{t1})^2 + (\sigma_{t2})^2 - \frac{2\ \sigma_{t1} \times \sigma_{t2}}{m}\right] = \frac{1}{2E}\left(\frac{\sigma_{yt}}{F.S.}\right)^2$$

or

$$(\sigma_{t1})^2 + (\sigma_{t2})^2 - \frac{2\ \sigma_{t1} \times \sigma_{t2}}{m} = \left(\frac{\sigma_{yt}}{F.S.}\right)^2$$

$$\left[\frac{211.9 \times 10^6}{d^3}\right]^2 + \left[\frac{-110.1 \times 10^6}{d^3}\right]^2 - 2 \times \frac{211.9 \times 10^6}{d^3} \times \frac{-110.1 \times 10^6}{d^3} \times 0.25 = \left(\frac{700}{2}\right)^2$$

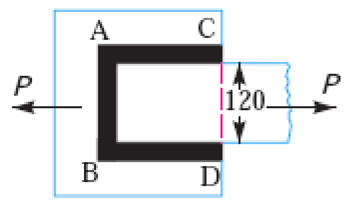
or

$$\frac{44\ 902 \times 10^{12}}{d^6} + \frac{12\ 122 \times 10^{12}}{d^6} + \frac{11\ 665 \times 10^{12}}{d^6} = 122\ 500$$

$$\frac{68\ 689 \times 10^{12}}{d^6} = 122\ 500$$

$$\therefore \quad d^6 = 68\ 689 \times 10^{12}/122\ 500 = 0.5607 \times 10^{12} \text{ or } d = 90.8 \text{ mm } \textbf{Ans.}$$

12. Determine the length of the weld run for a plate of size 120 mm wide and 15 mm thick to be welded to another plate by means of 1. A single transverse weld; and 2. Double parallel fillet welds when the joint is subjected to variable loads.



Given : Width = 120 mm ; Thickness = 15 mm In Fig. *AB* represents the single transverse weld and *AC* and *BD* represents double parallel fillet welds.

1. *Length of the weld run for a single transverse weld*

The effective length of the weld run (l_1) for a single transverse weld may be obtained by subtracting 12.5 mm from the width of the plate.

$$\therefore \quad l_1 = 120 - 12.5 = 107.5 \text{ mm } \textbf{Ans.}$$

2. *Length of the weld run for a double parallel fillet weld subjected to variable loads*

Let l_2 = Length of weld run for each parallel fillet, and

s = Size of weld = Thickness of plate = 15 mm

Assuming the tensile stress as 70 MPa or N/mm^2 and shear stress as 56 MPa or N/mm^2 for static loading. We know that the maximum load which the plate can carry is

$$P = \text{Area} \times \text{Stress} = 120 \times 15 \times 70 = 126 \times 10^3 \text{ N}$$

From Table 10.6, we find that the stress concentration factor for transverse weld is 1.5 and for parallel fillet welds is 2.7.

$\therefore$ Permissible tensile stress,

$$\sigma_t = 70 / 1.5 = 46.7 \text{ N/mm}^2$$

and permissible shear stress,

$$\tau = 56 / 2.7 = 20.74 \text{ N/mm}^2$$

∴ Load carried by single transverse weld,

$$P_1 = 0.707\, s \times l_1 \times \sigma_t = 0.707 \times 15 \times 107.5 \times 46.7 = 53\,240 \text{ N}$$

and load carried by double parallel fillet weld,

$$P_2 = 1.414\, s \times l_2 \times \tau = 1.414 \times 15 \times l_2 \times 20.74 = 440\, l_2 \text{ N}$$

∴ Load carried by the joint (P),

$$126 \times 10^3 = P_1 + P_2 = 53\,240 + 440\, l_2 \quad \text{or} \quad l_2 = 165.4 \text{ mm}$$

Adding 12.5 mm for starting and stopping of weld run, we have

$$l_2 = 165.4 + 12.5 = 177.9 \text{ say } 178 \text{ mm}$$ **Ans.**

13. A vertical two start square threaded screw of a 100 mm mean diameter and 20 mm pitch supports a vertical load of 18 kN. The axial thrust on the screw is taken by a collar bearing of 250 mm outside diameter and 100 mm inside diameter. Find the force required at the end of a lever which is 400 mm long in order to lift and lowe r the load. The coefficient of friction for the vertical screw and nut is 0.15 and that for collar bearing is 0.20

Solution. Given : $d = 100$ mm ; $p = 20$ mm ; $W = 18$ kN $= 18 \times 10^3$N ; $D_1 = 250$ mm or $R_1 = 125$ mm ; $D_2 = 100$ mm or $R_2 = 50$ mm ; $l = 400$ mm ; $\mu = \tan\phi = 0.15$; $\mu_1 = 0.20$

Force required at the end of lever

Let P = Force required at the end of lever.

Since the screw is a two start square threaded screw, therefore lead of the screw

$$= 2\,p = 2 \times 20 = 40 \text{ mm}$$

We know that $\tan\alpha = \dfrac{\text{Lead}}{\pi d} = \dfrac{40}{\pi \times 100} = 0.127$

1. ***For raising the load***

We know that tangential force required at the circumference of the screw,

$$P = W \tan(\alpha + \phi) = W\left[\frac{\tan\alpha + \tan\phi}{1 - \tan\alpha \tan\phi}\right]$$

$$= 18 \times 10^3 \left[\frac{0.127 + 0.15}{1 - 0.127 \times 0.15}\right] = 5083 \text{ N}$$

and mean radius of the collar,

$$R = \frac{R_1 + R_2}{2} = \frac{125 + 50}{2} = 87.5 \text{ mm}$$

∴ Total torque required at the end of lever,

$$T = P \times \frac{d}{2} + \mu_1 WR$$

$$= 5083 \times \frac{100}{2} + 0.20 \times 18 \times 10^3 \times 87.5 = 569\,150 \text{ N-mm}$$

We know that torque required at the end of lever (T),

$$569\,150 = P_1 \times l = P_1 \times 400 \quad \text{or} \quad P_1 = 569\,150/400 = 1423 \text{ N}$$ **Ans.**

2. *For lowering the load*

We know that tangential force required at the circumference of the screw,

$$P = W \tan(\phi - \alpha) = W\left[\frac{\tan\phi - \tan\alpha}{1 + \tan\phi \tan\alpha}\right]$$

$$= 18 \times 10^3 \left[\frac{0.15 - 0.127}{1 + 0.15 \times 0.127}\right] = 406.3 \text{ N}$$

and the total torque required the end of lever,

$$T = P \times \frac{d}{2} + \mu_1 W R$$

$$= 406.3 \times \frac{100}{2} + 0.20 \times 18 \times 10^3 \times 87.5 = 335\,315 \text{ N-mm}$$

We know that torque required at the end of lever (T),

$$335\,315 = P_1 \times l = P_1 \times 400 \quad \text{or} \quad P_1 = 335\,315 / 400 = 838.3 \text{ N} \textbf{ Ans.}$$

14. A rectangular strut is 150 mm wide and 120 mm thick. It carries a load of 180 kN at an eccentricity of 10 mm in a plane bisecting the thickness as shown in Fig. Find the maximum and minimum intensities of stress in the section.

Solution. Given : b = 150 mm ; d = 120 mm ; P = 180 kN = 180×10^3 N ; e = 10 mm

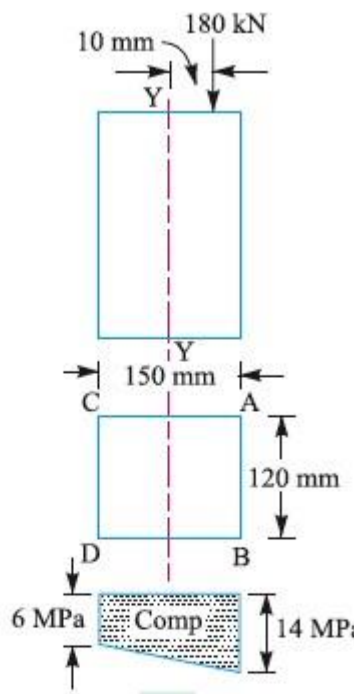


Fig.

We know that cross-sectional area of the strut,

$$A = b.d = 150 \times 120$$
$$= 18 \times 10^3 \text{ mm}^2$$

∴ Direct compressive stress,

$$\sigma_o = \frac{P}{A} = \frac{180 \times 10^3}{18 \times 10^3}$$
$$= 10 \text{ N/mm}^2 = 10 \text{ MPa}$$

Section modulus for the strut,

$$Z = \frac{I_{YY}}{y} = \frac{d \cdot b^3/12}{b/2} = \frac{d \cdot b^2}{6}$$
$$= \frac{120\,(150)^2}{6}$$
$$= 450 \times 10^3 \text{ mm}^3$$

Bending moment, $M = P.e = 180 \times 10^3 \times 10$
$= 1.8 \times 10^6$ N-mm

∴ Bending stress, $\sigma_b = \dfrac{M}{Z} = \dfrac{1.8 \times 10^6}{450 \times 10^3}$
$= 4 \text{ N/mm}^2 = 4$ MPa

Since σ_o is greater than σ_b, therefore the entire cross-section of the strut will be subjected to compressive stress. The maximum intensity of compressive stress will be at the edge AB and minimum at the edge CD.

∴ Maximum intensity of compressive stress at the edge AB

$$= \sigma_o + \sigma_b = 10 + 4 = 14 \text{ MPa} \textbf{ Ans.}$$

and minimum intensity of compressive stress at the edge CD

$$= \sigma_o - \sigma_b = 10 - 4 = 6 \text{ MPa} \textbf{ Ans.}$$

15. Design and draw a cast iron flange coupling for a mild steel shaft transmitting 90 kW at 250 r.p.m. The allowable shear stress in the shaft is 40 MPa and the angle of twist is not to exceed 1° in a length of 20 diameters. The allowable shear stress in the coupling bolts is 30 MPa.

Solution. Given : $P = 90$ kW $= 90 \times 10^3$ W ; $N = 250$ r.p.m. ; $\tau_s = 40$ MPa $= 40$ N/mm^2 ; $\theta = 1° = \pi / 180 = 0.0175$ rad ; $\tau_b = 30$ MPa $= 30$ N/mm^2

First of all, let us find the diameter of the shaft (d). We know that the torque transmitted by the shaft,

$$T = \frac{P \times 60}{2\pi N} = \frac{90 \times 10^3 \times 60}{2\pi \times 250} = 3440 \text{ N-m} = 3440 \times 10^3 \text{ N-mm}$$

Considering strength of the shaft, we know that

$$\frac{T}{J} = \frac{\tau_s}{d/2}$$

$$\frac{3440 \times 10^3}{\frac{\pi}{32} \times d^4} = \frac{40}{d/2} \quad \text{or} \quad \frac{35 \times 10^6}{d^4} = \frac{80}{d} \qquad \ldots \left(\because J = \frac{\pi}{32} \times d^4\right)$$

$$\therefore \quad d^3 = 35 \times 10^6 / 80 = 0.438 \times 10^6 \quad \text{or} \quad d = 76 \text{ mm}$$

Considering rigidity of the shaft, we know that

$$\frac{T}{J} = \frac{C \times \theta}{l}$$

$$\frac{3440 \times 10^3}{\frac{\pi}{32} \times d^4} = \frac{84 \times 10^3 \times 0.0175}{20\,d} \quad \text{or} \quad \frac{35 \times 10^6}{d^4} = \frac{73.5}{d} \qquad \ldots (\text{Taking } C = 84 \text{ kN/mm}^2)$$

$$\therefore \quad d^3 = 35 \times 10^6 / 73.5 = 0.476 \times 10^6 \quad \text{or} \quad d = 78 \text{ mm}$$

Taking the larger of the two values, we have

$d = 78$ say 80 mm **Ans.**

Let us now design the cast iron flange coupling of the protective type as discussed below :

1. *Design for hub*

We know that the outer diameter of hub,

$D = 2d = 2 \times 80 = 160$ mm **Ans.**

and length of hub, $L = 1.5\,d = 1.5 \times 80 = 120$ mm **Ans.**

Let us now check the induced shear stress in the hub by considering it as a hollow shaft. The shear stress for the hub material (which is cast iron) is usually 14 MPa. We know that the torque transmitted (T),

$$3440 \times 10^3 = \frac{\pi}{16} \times \tau_c \left[\frac{D^4 - d^4}{D}\right] = \frac{\pi}{16} \times \tau_c \left[\frac{(160)^4 - (80)^4}{160}\right] = 754 \times 10^3\, \tau_c$$

$$\therefore \quad \tau_c = 3440 \times 10^3 / 754 \times 10^3 = 4.56 \text{ N/mm}^2 = 4.56 \text{ MPa}$$

Since the induced shear stress for the hub material is less than 14 MPa, therefore the design for hub is safe.

2. *Design for key*

From Table 13.1, we find that the proportions of key for a 80 mm diameter shaft are :

Width of key, $w = 25$ mm **Ans.**

and thickness of key, $t = 14$ mm **Ans.**

The length of key (l) is taken equal to the length of hub (L).

$\therefore \quad l = L = 120$ mm **Ans.**

Assuming that the shaft and key are of the same material. Let us now check the induced shear stress in key. We know that the torque transmitted (T),

$$3440 \times 10^3 = l \times w \times \tau_k \times \frac{d}{2} = 120 \times 25 \times \tau_k \times \frac{80}{2} = 120 \times 10^3\ \tau_k$$

$$\tau_k = 3440 \times 10^3/120 \times 10^3 = 28.7 \text{ N/mm}^2 = 28.7 \text{ MPa}$$

Since the induced shear stress in the key is less than 40 MPa, therefore the design for key is safe.

3. Design for flange

The thickness of the flange (t_f) is taken as 0.5 d.

$\therefore \quad t_f = 0.5\, d = 0.5 \times 80 = 40$ mm **Ans.**

Let us now check the induced shear stress in the cast iron flange by considering the flange at the junction of the hub under shear. We know that the torque transmitted (T),

$$3440 \times 10^3 = \frac{\pi D^2}{2} \times t_f \times \tau_c = \frac{\pi (160)^2}{2} \times 40 \times \tau_c = 1608 \times 10^3\ \tau_c$$

$$\therefore \quad \tau_c = 3440 \times 10^3/1608 \times 10^3 = 2.14 \text{ N/mm}^2 = 2.14 \text{ MPa}$$

Since the induced shear stress in the flange is less than 14 MPa, therefore the design for flange is safe.

4. Design for bolts

Let $\quad d_1$ = Nominal diameter of bolts.

Since the diameter of the shaft is 80 mm, therefore let us take number of bolts,

$$n = 4$$

and pitch circle diameter of bolts,

$$D_1 = 3\, d = 3 \times 80 = 240 \text{ mm}$$

The bolts are subjected to shear stress due to the torque transmitted. We know that torque transmitted (T),

$$3440 \times 10^3 = \frac{\pi}{4} (d_1)^2\, n \times \tau_b \times \frac{D_1}{2} = \frac{\pi}{4} (d_1)^2 \times 4 \times 30 \times \frac{240}{2} = 11\ 311\ (d_1)^2$$

$$\therefore \quad (d_1)^2 = 3440 \times 10^3/11\ 311 = 304 \text{ or } d_1 = 17.4 \text{ mm}$$

Assuming coarse threads, the standard nominal diameter of bolt is 18 mm. **Ans.**

The other proportions are taken as follows :

Outer diameter of the flange,

$D_2 = 4\, d = 4 \times 80 = 320$ mm **Ans.**

Thickness of protective circumferential flange,

$t_p = 0.25\, d = 0.25 \times 80 = 20$ mm **Ans.**

16. An engine developing 45 kW at 1000 r.p.m. is fitted with a cone clutch built inside the flywheel. The cone has a face angle of 12.5° and a maximum mean diameter of 500 mm. The coefficient of friction is 0.2. The normal pressure on the clutch face is not to exceed 0.1 N/mm2. Determine: 1. the face width required, and 2. the axial spring force necessary to engage the clutch.

Solution. Given : $P = 45$ kW $= 45 \times 10^3$ W ; $N = 1000$ r.p.m. ; $\alpha = 12.5°$; $D = 500$ mm or $R = 250$ mm ; $\mu = 0.2$; $p_n = 0.1$ N/mm^2

1. Face width

Let b = Face width of the clutch in mm.

We know that torque developed by the clutch,

$$T = \frac{P \times 60}{2\,\pi\,N} = \frac{45 \times 10^3 \times 60}{2\,\pi \times 1000} = 430 \text{ N-m} = 430 \times 10^3 \text{ N-mm}$$

We also know that torque developed by the clutch (T),

$$430 \times 10^3 = 2\pi.\ \mu.\ p_n.\ R^2.b = 2\pi \times 0.2 \times 0.1\ (250)^2 b = 7855\ b$$

$$\therefore \quad b = 430 \times 10^3 / 7855 = 54.7 \text{ say } 55 \text{ mm } \textbf{Ans.}$$

2. Axial spring force necessary to engage the clutch

We know that the normal force acting on the contact surfaces,

$$W_n = p_n \times 2\pi R.b = 0.1 \times 2\pi \times 250 \times 55 = 8640 \text{ N}$$

$\therefore$ Axial spring force necessary to engage the clutch,

$$W_e = W_n\,(\sin\alpha + 0.25\,\mu\cos\alpha)$$

$$= 8640\,(\sin 12.5° + 0.25 \times 0.2 \cos 12.5°) = 2290 \text{ N } \textbf{Ans.}$$

17. Design a spring for a balance to measure 0 to 1000 N over a scale of length 80 mm. The spring is to be enclosed in a casing of 25 mm diameter. The approximate number of turns is 30. The modulus of rigidity is 85 kN/mm^2. Also calculate the maximum shear stress induced.

Solution. Given: $W = 1000$ N; $\delta = 80$ mm; n = 30; $G = 85$ kN/mm^2 $= 85 \times 10^3$ N/mm^2

Design of spring

Let D = Mean diameter of the spring coil,

d = Diameter of the spring wire, and

C = Spring index = D/d.

Since the spring is to be enclosed in a casing of 25 mm diameter, therefore the outer diameter of the spring coil ($D_o = D + d$) should be less than 25 mm.

We know that deflection of the spring (δ),

$$80 = \frac{8\,W.C^3.n}{G.d} = \frac{8 \times 1000 \times C^3 \times 30}{85 \times 10^3 \times d} = \frac{240\,C^3}{85\,d}$$

$$\therefore \quad \frac{C^3}{d} = \frac{80 \times 85}{240} = 28.3$$

Let us assume that $d = 4$ mm. Therefore

$$C^3 = 28.3\,d = 28.3 \times 4 = 113.2 \quad \text{or} \quad C = 4.84$$

and $D = C.d = 4.84 \times 4 = 19.36$ mm **Ans.**

We know that outer diameter of the spring coil,

$$D_o = D + d = 19.36 + 4 = 23.36 \text{ mm } \textbf{Ans.}$$

Since the value of $D_o = 23.36$ mm is less than the casing diameter of 25 mm, therefore the assumed dimension, $d = 4$ mm is correct.

Maximum shear stress induced

We know that Wahl's stress factor,

$$K = \frac{4C-1}{4C-4} + \frac{0.615}{C} = \frac{4 \times 4.84 - 1}{4 \times 4.84 - 4} + \frac{0.615}{4.84} = 1.322$$

∴ Maximum shear stress induced,

$$\tau = K \times \frac{8\,W.C}{\pi\,d^2} = 1.322 \times \frac{8 \times 1000 \times 4.84}{\pi \times 4^2}$$
$$= 1018.2 \text{ N/mm}^2 = 1018.2 \text{ MPa} \textbf{ Ans.}$$

18. **Design and draw a cotter joint to support a load varying from 30 kN in compression to 30 kN in tension. The material used is carbon steel for which the following allowable stresses may be used. The load is applied statically. Tensile stress = compressive stress = 50 MPa; shear stress = 35 MPa and crushing stress = 90 MPa.**

Solution. Given: P = 30 kN = 30 × 10^3 N; σ_t = 50 MPa = 50 N / mm^2; τ = 35 MPa = 35 N/ mm^2; σ_c = 90 MPa = 90 N/mm^2

1. Diameter of the rods

Let d = Diameter of the rods.

Considering the failure of the rod in tension. We know that load (P),

$$30 \times 10^3 = \frac{\pi}{4} \times d^2 \times \sigma_t = \frac{\pi}{4} \times d^2 \times 50 = 39.3\,d^2$$

$$\therefore \quad d^2 = 30 \times 10^3 / 39.3 = 763 \quad \text{or} \quad d = 27.6 \text{ say } 28 \text{ mm} \quad \textbf{Ans.}$$

2. Diameter of spigot and thickness of cotter

Let d2 = Diameter of spigot or inside diameter of socket, and

t = Thickness of cotter. It may be taken as d2 / 4.

Let d_2 = Diameter of spigot or inside diameter of socket, and

t = Thickness of cotter. It may be taken as $d_2 / 4$.

Considering the failure of spigot in tension across the weakest section. We know that load (P),

$$30 \times 10^3 = \left[\frac{\pi}{4}(d_2)^2 - d_2 \times t\right]\sigma_t = \left[\frac{\pi}{4}(d_2)^2 - d_2 \times \frac{d_2}{4}\right]50 = 26.8\,(d_2)^2$$

$$\therefore \quad (d_2)^2 = 30 \times 10^3 / 26.8 = 1119.4 \quad \text{or} \quad d_2 = 33.4 \text{ say } 34 \text{ mm}$$

and thickness of cotter, $t = \frac{d_2}{4} = \frac{34}{4} = 8.5 \text{ mm}$

Let us now check the induced crushing stress. We know that load (P),

$$30 \times 10^3 = d_2 \times t \times \sigma_c = 34 \times 8.5 \times \sigma_c = 289\,\sigma_c$$

$$\therefore \quad \sigma_c = 30 \times 10^3 / 289 = 103.8 \text{ N/mm}^2$$

Since this value of σ_c is more than the given value of $\sigma_c = 90$ N/mm², therefore the dimensions d_2 = 34 mm and t = 8.5 mm are not safe. Now let us find the values of d_2 and t by substituting the value of $\sigma_c = 90$ N/mm² in the above expression, *i.e.*

$$30 \times 10^3 = d_2 \times \frac{d_2}{4} \times 90 = 22.5\,(d_2)^2$$

$$\therefore \quad (d_2)^2 = 30 \times 10^3 / 22.5 = 1333 \quad \text{or} \quad d_2 = 36.5 \text{ say } 40 \text{ mm} \quad \textbf{Ans.}$$

and $t = d_2 / 4 = 40 / 4 = 10 \text{ mm}$ **Ans.**

3. *Outside diameter of socket*

Let d_1 = Outside diameter of socket.

Considering the failure of the socket in tension across the slot. We know that load (P),

$$30 \times 10^3 = \left[\frac{\pi}{4}\left\{(d_1)^2 - (d_2)^2\right\} - (d_1 - d_2)\, t\right] \sigma_t$$

$$= \left[\frac{\pi}{4}\left\{(d_1)^2 - (40)^2\right\} - (d_1 - 40)\, 10\right] 50$$

$$30 \times 10^3/50 = 0.7854\,(d_1)^2 - 1256.6 - 10\, d_1 + 400$$

or $(d_1)^2 - 12.7\, d_1 - 1854.6 = 0$

$$\therefore \quad d_1 = \frac{12.7 \pm \sqrt{(12.7)^2 + 4 \times 1854.6}}{2} = \frac{12.7 \pm 87.1}{2}$$

$= 49.9$ say 50 mm **Ans.** ...(Taking +ve sign)

4. *Width of cotter*

Let b = Width of cotter.

Considering the failure of the cotter in shear. Since the cotter is in double shear, therefore load (P),

$$30 \times 10^3 = 2\, b \times t \times \tau = 2\, b \times 10 \times 35 = 700\, b$$

$$\therefore \quad b = 30 \times 10^3 / 700 = 43 \text{ mm}$$ **Ans.**

5. *Diameter of socket collar*

Let d_4 = Diameter of socket collar.

Considering the failure of the socket collar and cotter in crushing. We know that load (P),

$$30 \times 10^3 = (d_4 - d_2)\, t \times \sigma_c = (d_4 - 40) 10 \times 90 = (d_4 - 40)\, 900$$

$$\therefore \quad d_4 - 40 = 30 \times 10^3 / 900 = 33.3 \quad \text{or} \quad d_4 = 33.3 + 40 = 73.3 \text{ say } 75 \text{ mm}$$ **Ans.**

6. *Thickness of socket collar*

Let c = Thickness of socket collar.

Considering the failure of the socket end in shearing. Since the socket end is in double shear, therefore load (P),

$$30 \times 10^3 = 2(d_4 - d_2)\, c \times \tau = 2\,(75 - 40)\, c \times 35 = 2450\, c$$

$$\therefore \quad c = 30 \times 10^3 / 2450 = 12 \text{ mm}$$ **Ans.**

7. *Distance from the end of the slot to the end of the rod*

Let a = Distance from the end of slot to the end of the rod.

Considering the failure of the rod end in shear. Since the rod end is in double shear, therefore load (P),

$$30 \times 10^3 = 2\, a \times d_2 \times \tau = 2a \times 40 \times 35 = 2800\, a$$

$$\therefore \quad a = 30 \times 10^3 / 2800 = 10.7 \text{ say } 11 \text{ mm}$$ **Ans.**

8. *Diameter of spigot collar*

Let d_3 = Diameter of spigot collar.

Considering the failure of spigot collar in crushing. We know that load (P),

$$30 \times 10^3 = \frac{\pi}{4}\left[(d_3)^2 - (d_2)^2\right] \sigma_c = \frac{\pi}{4}\left[(d_3)^2 - (40)^2\right] 90$$

or
$$(d_3)^2 - (40)^2 = \frac{30 \times 10^3 \times 4}{90 \times \pi} = 424$$

$\therefore \quad (d_3)^2 = 424 + (40)^2 = 2024 \quad \text{or} \quad d_3 = 45 \text{ mm}$ **Ans.**

9. Thickness of spigot collar

Let t_1 = Thickness of spigot collar.

Considering the failure of spigot collar in shearing. We know that load (P)

$$30 \times 10^3 = \pi\, d_2 \times t_1 \times \tau = \pi \times 40 \times t_1 \times 35 = 4400\, t_1$$

$\therefore \quad t_1 = 30 \times 10^3 / 4400 = 6.8$ say 8 mm **Ans.**

10. The length of cotter (l) is taken as 4 d.

$\therefore \quad l = 4\,d = 4 \times 28 = 112$ mm **Ans.**

11. The dimension e is taken as 1.2 d.

$\therefore \quad e = 1.2 \times 28 = 33.6$ say 34 mm **Ans.**

19. A machine component is subjected to a flexural stress which fluctuates between + 300 MN/m2 and – 150 MN/m2. Determine the value of minimum ultimate strength according to Modified Goodman relation; and Soderberg relation. Take yield strength = 0.55 Ultimate strength; Endurance strength = 0.5 Ultimate strength; and factor of safety = 2.

Solution. Given : $\sigma_1 = 300$ MN/m^2 ; $\sigma_2 = -150$ MN/m^2 ; $\sigma_y = 0.55\,\sigma_u$; $\sigma_e = 0.5\,\sigma_u$; $F.S. = 2$

Let σ_u = Minimum ultimate strength in MN/m^2.

We know that the mean or average stress,

$$\sigma_m = \frac{\sigma_1 + \sigma_2}{2} = \frac{300 + (-150)}{2} = 75 \text{ MN/m}^2$$

and variable stress,

$$\sigma_v = \frac{\sigma_1 - \sigma_2}{2} = \frac{300 - (-150)}{2} = 225 \text{ MN/m}^2$$

According to modified Goodman relation

We know that according to modified Goodman relation,

$$\frac{1}{F.S.} = \frac{\sigma_m}{\sigma_u} + \frac{\sigma_v}{\sigma_e}$$

or

$$\frac{1}{2} = \frac{75}{\sigma_u} + \frac{225}{0.5\,\sigma_u} = \frac{525}{\sigma_u}$$

$\therefore \quad \sigma_u = 2 \times 525 = 1050$ MN/m^2 **Ans.**

According to Soderberg relation

We know that according to Soderberg relation,

$$\frac{1}{F.S.} = \frac{\sigma_m}{\sigma_y} + \frac{\sigma_v}{\sigma_e}$$

or

$$\frac{1}{2} = \frac{75}{0.55\,\sigma_u} + \frac{255}{0.5\,\sigma_u} = \frac{586.36}{\sigma_u}$$

$\therefore \quad \sigma_u = 2 \times 586.36 = 1172.72$ MN/m^2 **Ans.**

20. Determine the thickness of a 120 mm wide uniform plate for safe continuous operation if the plate is to be subjected to a tensile load that has a maximum value of 250 kN and a minimum value of 100 kN. The properties of the plate material are as follows: Endurance limit stress = 225 MPa, and Yield point stress = 300 MPa. The factor of safety based on yield point may be taken as 1.5.

Solution. Given : $b = 120$ mm ; $W_{max} = 250$ kN; $W_{min} = 100$ kN ; $\sigma_e = 225$ MPa $= 225$ N/mm^2 ; $\sigma_y = 300$ MPa $= 300$ N/mm^2; $F.S. = 1.5$

Let t = Thickness of the plate in mm.

$\therefore$ Area, $A = b \times t = 120\,t$ mm^2

We know that mean or average load,

$$W_m = \frac{W_{max} + W_{min}}{2} = \frac{250 + 100}{2} = 175 \text{ kN} = 175 \times 10^3 \text{ N}$$

$$\therefore \quad \text{Mean stress, } \sigma_m = \frac{W_m}{A} = \frac{175 \times 10^3}{120\,t} \text{ N/mm}^2$$

$$\text{Variable load, } W_v = \frac{W_{max} - W_{min}}{2} = \frac{250 - 100}{2} = 75 \text{ kN} = 75 \times 10^3 \text{ N}$$

$$\therefore \quad \text{Variable stress, } \sigma_v = \frac{W_v}{A} = \frac{75 \times 10^3}{120\,t} \text{ N/mm}^2$$

According to Soderberg's formula,

$$\frac{1}{F.S.} = \frac{\sigma_m}{\sigma_y} + \frac{\sigma_v}{\sigma_e}$$

$$\frac{1}{1.5} = \frac{175 \times 10^3}{120\,t \times 300} + \frac{75 \times 10^3}{120\,t \times 225} = \frac{4.86}{t} + \frac{2.78}{t} = \frac{7.64}{t}$$

$\therefore$ $t = 7.64 \times 1.5 = 11.46$ say 11.5 mm **Ans.**

5615121

B.TECH. DEGREE EXAMINATION APRIL/MAY 2016

Fifth semester – Mechanical Engineering

DESIGN OF MACHINE ELEMENTS

(From 2013-14 batches onwards)

Part A

1. **What are the various phases of design process?**
   - ✓ Recognition of need
   - ✓ Definition of problem
   - ✓ Synthesis
   - ✓ Analysis and Optimization
   - ✓ Evaluation
   - ✓ Presentation
2. **What are the practical applications of welded joints?**

   The practical applications of welded joints are machine frame, bases, jigs, fixtures; it is also used in high pressure steam pipe, aircraft, automotive and ship buildings industries.
3. **Define factor of safety.**

   Factor of safety (FOS) is defined as the ratio between the ultimate stress and working stress.
4. **What are the different types of loads that can act on machine components?**
   - ✓ Dead or steady load.
   - ✓ Live or variable load.
   - ✓ Suddenly applied or shock loads.
   - ✓ Impact load.
5. **What is eccentric load and eccentricity?**

   An external load, whose line of action is parallel but does not coincide with the centroidal axis of the machine component, is known as an eccentric load.

   The distance between the centroidal axis of the machine component and the eccentric load is called eccentricity
6. **Name any two of the rigid and flexible couplings.**

   Rigid coupling.
   - ✓ Sleeve or muff coupling.
   - ✓ Flange coupling.

   Flexible coupling.
   - ✓ Bushed pin type coupling,
   - ✓ Universal coupling, and
7. **What is curved beam? Give some examples for curved beam?**

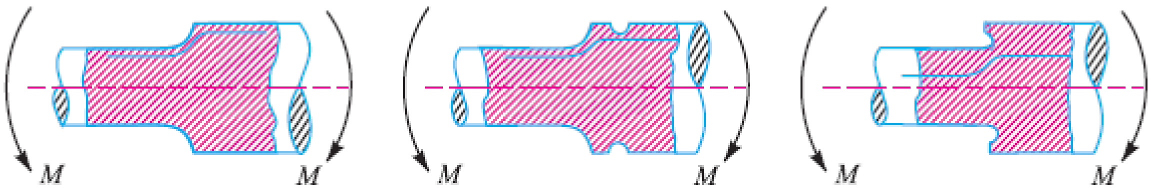
   The neutral axis of the cross-section is shifted towards the centre of curvature of the beam causing a non-linear distribution of stress.

   The application of curved beam principle is used in crane hooks, chain links and frames of punches, presses, and planers.
8. **Define "Spring Index".**

   It is the ratio of mean pitch diameter to the diameter of the wire(C= D/d).
9. **Differentiate between shaft and axle.**

   Shat is nothing but a rotating machine element which transmits power. An axle, through similar in shape to the shaft, is a stationary machine element and is used for transmission of bending moment only. It simply acts as a support for some rotating body.
10. **Give some methods of Reducing Stress Concentration.**

- ✓ Methods of reducing stress concentration in cylindrical members with shoulders



(*a*) Poor (*b*) Good (*c*) Preferred

- ✓ Methods of reducing stress concentration in cylindrical members with holes.

(*a*) Poor (*b*) Preferred

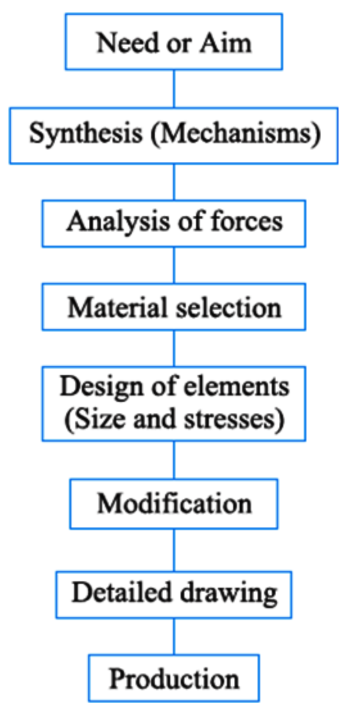
Part B(5 x 11=55)

Unit I

11. a) Give the General Procedure in Machine Design

In designing a machine component, there is no rigid rule. The problem may be attempted in several ways. However, the general procedure to solve a design problem is as follows:

- ✓ **Recognition of need.** First of all, make a complete statement of the problem, indicating the need, aim or purpose for which the machine is to be designed.
- ✓ **Synthesis (Mechanisms).** Select the possible mechanism or group of mechanisms which will give the desired motion.
- ✓ **Analysis of forces.** Find the forces acting on each member of the machine and the energy transmitted by each member.
- ✓ **Material selection.** Select the material best suited for each member of the machine.



General Procedure in Machine Design

- ✓ **Design of elements (Size and Stresses).** Find the size of each member of the machine by considering the force acting on the member and the permissible stresses for the material used.

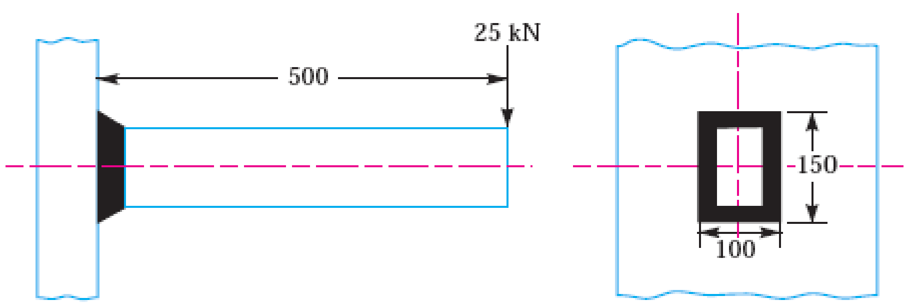
It should be kept in mind that each member should not deflect or deform than the permissible limit.

- ✓ **Modification.** Modify the size of the member to agree with the past experience and judgment to facilitate manufacture. The modification may also be necessary by consideration of manufacturing to reduce overall cost.
- ✓ **Detailed drawing.** Draw the detailed drawing of each component and the assembly of the machine with complete specification for the manufacturing processes suggested.
- ✓ **Production.** The component, as per the drawing, is manufactured in the workshop.

b) What are the General Considerations in Machine Design?

- ✓ Type of load and stresses caused by the load
- ✓ Motion of the parts or kinematics of the machine
- ✓ Selection of materials
- ✓ Form and size of the parts.
- ✓ Frictional resistance and lubrication
- ✓ Convenient and economical features
- ✓ Use of standard parts
- ✓ Safety of operation
- ✓ Workshop facilities
- ✓ Cost of construction
- ✓ Assembling

12. A rectangular cross-section bar is welded to a support by means of fillet welds as shown in Fig. Determine the size of the welds, if the permissible shear stress in the weld is limited to 75 MPa.



All dimensions in mm

Given: $P = 25$ kN $= 25 \times 10^3$N; $\tau_{max} = 75$ MPa $= 75$N/mm^2; $l = 100$ mm; $b = 150$ mm; $e = 500$ mm

Let s = Size of the weld, and

t = Throat thickness.

The joint, as shown in Fig, is subjected to direct shear stress and the bending stress. We know that the throat area for a rectangular fillet weld,

$$A = t\,(2b + 2l) = 0.707\, s\,(2b + 2l)$$

$$= 0.707s\,(2 \times 150 + 2 \times 100) = 353.5\, s \text{ mm}^2 \qquad \ldots (\because\ t = 0.707$$

$$\therefore \text{ Direct shear stress, } \tau = \frac{P}{A} = \frac{25 \times 10^3}{353.5\, s} = \frac{70.72}{s} \text{ N/mm}^2$$

We know that bending moment,

$$M = P \times e = 25 \times 10^3 \times 500 = 12.5 \times 10^6 \text{ N-mm}$$

From Table 10.7, we find that for a rectangular section, section modulus,

$$Z = t\left(b.l + \frac{b^2}{3}\right) = 0.707\ s\left[150 \times 100 + \frac{(150)^2}{3}\right] = 15\ 907.5\ s\ \text{mm}$$

$\therefore$ Bending stress, $$\sigma_b = \frac{M}{Z} = \frac{12.5 \times 10^6}{15\ 907.5\, s} = \frac{785.8}{s}\ \text{N/mm}^2$$

We know that maximum shear stress (τ_{max}),

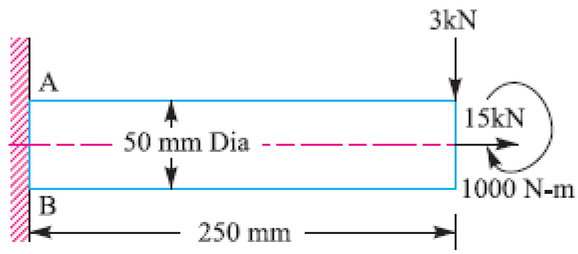
$$75 = \frac{1}{2}\sqrt{(\sigma_b)^2 + 4\,\tau^2} = \frac{1}{2}\sqrt{\left(\frac{785.8}{s}\right)^2 + 4\left(\frac{70.72}{s}\right)^2} = \frac{399.2}{s}$$

$$s = 399.2 / 75 = 5.32 \text{ mm } \textbf{Ans.}$$

<u>Unit II</u>

13. A shaft, as shown in Fig. 5.17, is subjected to a bending load of 3 kN, pure torqueof 1000 N-m and an axial pulling force of 15 kN. Calculate the stresses at A and B. All dimensions are in mm.

Given: W = 3 kN = 3000 N;T = 1000 N-m = 1 × 10^6 N-mm; P = 15 kN= 15 × 103 N; d = 50 mm; x = 250 mm



We know that cross-sectional area of the shaft

$$A = \frac{\pi}{4} \times d^2$$

$$= \frac{\pi}{4}(50)^2 = 1964 \text{ mm}^2$$

$\therefore$Tensile stress due to axial pulling at points A and B,

$$\sigma_o = \frac{P}{A} = \frac{15 \times 10^3}{1964} = 7.64 \text{ N/mm}^2 = 7.64 \text{ MPa}$$

Bending moment at points A and B,

$$M = W.x = 3000 \times 250 = 750 \times 10^3 \text{ N-mm}$$

Section modulus for the shaft,

$$Z = \frac{\pi}{32} \times d^3 = \frac{\pi}{32}(50)^3$$

$$= 12.27 \times 10^3 \text{ mm}^3$$

$\therefore$ Bending stress at points A and B,

$$\sigma_b = \frac{M}{Z} = \frac{750 \times 10^3}{12.27 \times 10^3}$$

$$= 61.1 \text{ N/mm}^2 = 61.1 \text{ MPa}$$

This bending stress is tensile at point A andcompressive at point B.

$\therefore$ Resultant tensile stress at point A,

$$\sigma_A = \sigma_b + \sigma_o = 61.1 + 7.64$$
$$= 68.74 \text{ MPa}$$

and resultant compressive stress at point *B*,

$$\sigma_B = \sigma_b - \sigma_o = 61.1 - 7.64 = 53.46 \text{ MPa}$$

We know that the shear stress at points *A* and *B* due to the torque transmitted,

$$\tau = \frac{16\,T}{\pi\,d^3} = \frac{16 \times 1 \times 10^6}{\pi\,(50)^3} = 40.74 \text{ N/mm}^2 = 40.74 \text{ MPa} \qquad \ldots\left(\because\ T = \frac{\pi}{16} \times \tau \times d^3\right)$$

Stresses at point A

We know that maximum principal (or normal) stress at point *A*,

$$\sigma_{A(max)} = \frac{\sigma_A}{2} + \frac{1}{2}\left[\sqrt{(\sigma_A)^2 + 4\,\tau^2}\right]$$

$$= \frac{68.74}{2} + \frac{1}{2}\left[\sqrt{(68.74)^2 + 4\,(40.74)^2}\right]$$

$$= 34.37 + 53.3 = 87.67 \text{ MPa (tensile)}$$

Minimum principal (or normal) stress at point *A*,

$$\sigma_{A(min)} = \frac{\sigma_A}{2} - \frac{1}{2}\left[\sqrt{(\sigma_A)^2 + 4\,\tau^2}\right] = 34.37 - 53.3 = -18.93 \text{ MPa}$$

$$= 18.93 \text{ MPa (compressive)}$$

and maximum shear stress at point *A*,

$$\tau_{A(max)} = \frac{1}{2}\left[\sqrt{(\sigma_A)^2 + 4\,\tau^2}\right] = \frac{1}{2}\left[\sqrt{(68.74)^2 + 4\,(40.74)^2}\right]$$

$$= 53.3 \text{ MPa}$$

Stresses at point B

We know that maximum principal (or normal) stress at point *B*,

$$\sigma_{B(max)} = \frac{\sigma_B}{2} + \frac{1}{2}\left[\sqrt{(\sigma_B)^2 + 4\,\tau^2}\right]$$

$$= \frac{53.46}{2} + \frac{1}{2}\left[\sqrt{(53.46)^2 + 4\,(40.74)^2}\right]$$

$$= 26.73 + 48.73 = 75.46 \text{ MPa (compressive)}$$

Minimum principal (or normal) stress at point *B*,

$$\sigma_{B(min)} = \frac{\sigma_B}{2} - \frac{1}{2}\left[\sqrt{(\sigma_B)^2 + 4\tau^2}\right]$$

$$= 26.73 - 48.73 = -22 \text{ MPa}$$

$$= 22 \text{ MPa (tensile)}$$

and maximum shear stress at point *B*,

$$\tau_{B(max)} = \frac{1}{2}\left[\sqrt{(\sigma_B)^2 + 4\,\tau^2}\right] = \frac{1}{2}\left[\sqrt{(53.46)^2 + 4\,(40.74)^2}\right]$$

$$= 48.73 \text{ MPa}$$

14. A mild steel bracket as shown in Fig. is subjected to a pull of 6000 N acting at 45° to its horizontal axis. The bracket has a rectangular section whose depth is twice the thickness. Find the cross-sectional dimensions of the bracket, if the permissible stress in the material of the bracket is limited to 60 MPa.

Solution. Given : $P = 6000$ N ; $\theta = 45°$; $\sigma = 60$ MPa $= 60$ N/mm^2

Let t = Thickness of the section in mm, and

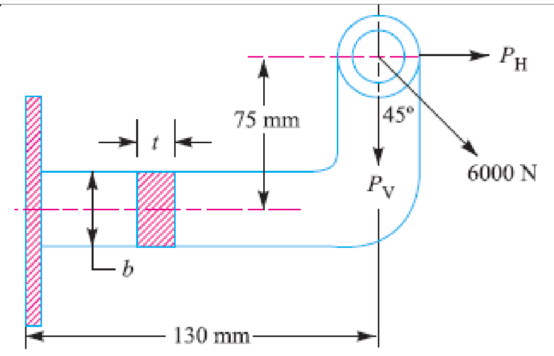
b = Depth or width of the section $= 2\,t$ ...(Given)

We know that area of cross-section,

$$A = b \times t = 2\,t \times t = 2\,t^2 \text{ mm}^2$$

and section modulus,

$$Z = \frac{t \times b^2}{6} = \frac{t\,(2t)^2}{6} = \frac{4\,t^3}{6} \text{ mm}^3$$



Horizontal component of the load,

$$P_H = 6000 \cos 45° = 6000 \times 0.707 = 4242 \text{ N}$$

$\therefore$ Bending moment due to horizontal component of the load,

$$M_H = P_H \times 75 = 4242 \times 75 = 318\,150 \text{ N-mm}$$

$\therefore$ Maximum bending stress on the upper surface due to horizontal omponent,

$$\sigma_{bH} = \frac{M_H}{Z} = \frac{318\,150 \times 6}{4\,t^3} = \frac{477\,225}{t^3} \text{ N/mm}^2 \text{ (tensile)}$$

Vertical component of the load,

$$P_V = 6000 \sin 45° = 6000 \times 0.707 = 4242 \text{ N}$$

$\therefore$ Direct stress due to vertical component,

$$\sigma_{oV} = \frac{P_V}{A} = \frac{4242}{2t^2} = \frac{2121}{t^2} \text{ N/mm}^2 \text{ (tensile)}$$

Bending moment due to vertical component of the load,

$$M_V = P_V \times 130 = 4242 \times 130 = 551\,460 \text{ N-mm}$$

This bending moment induces tensile stress on the upper surface and compressive stress on the lower surface of the bracket.

$\therefore$ Maximum bending stress on the upper surface due to vertical component,

$$\sigma_{bV} = \frac{M_V}{Z} = \frac{551\,460 \times 6}{4\,t^3} = \frac{827\,190}{t^3} \text{ N/mm}^2 \text{ (tensile)}$$

and total tensile stress on the upper surface of the bracket,

$$\sigma = \frac{477\,225}{t^3} + \frac{2121}{t^2} + \frac{827\,190}{t^3} = \frac{1\,304\,415}{t^3} + \frac{2121}{t^2}$$

Since the permissible stress (σ) is 60 N/mm^2, therefore

$$\frac{1\,304\,415}{t^3} + \frac{2121}{t^2} = 60 \text{ or } \frac{21\,740}{t^3} + \frac{35.4}{t^2} = 1$$

$\therefore$ $t = 28.4$ mm **Ans.** ... (By hit and trial)

and $b = 2t = 2 \times 28.4 = 56.8$ mm **Ans.**

Unit III

15. **Design a cast iron protective type flange coupling to transmit 15 kW at 900r.p.m. from an electric motor to a compressor. The service factor may be assumed as 1.35. The following permissible stresses may be used: Shear stress for shaft, bolt and key material = 40 MPa Crushing stress for bolt and key = 80 MPa, Shear stress for cast iron = 8 MPa, Draw a neat sketch of the coupling.**

Solution. Given : $P = 15$ kW $= 15 \times 10^3$ W ; $N = 900$ r.p.m. ; Service factor = 1.35 ; $\tau_s = \tau_b = \tau_k = 40$ MPa = 40 N/mm^2; $\sigma_{cb} = \sigma_{ck} = 80$ MPa = 80 N/mm^2 ; $\tau_c = 8$ MPa = 8 N/mm^2

The protective type flange coupling is designed as discussed below :

1. *Design for hub*

First of all, let us find the diameter of the shaft (d). We know that the torque transmitted by the shaft,

$$T = \frac{P \times 60}{2\,\pi N} = \frac{15 \times 10^3 \times 60}{2\,\pi \times 900} = 159.13 \text{ N-m}$$

Since the service factor is 1.35, therefore the maximum torque transmitted by the shaft,

$$T_{max} = 1.35 \times 159.13 = 215 \text{ N-m} = 215 \times 10^3 \text{ N-mm}$$

We know that the torque transmitted by the shaft (T),

$$215 \times 10^3 = \frac{\pi}{16} \times \tau_s \times d^3 = \frac{\pi}{16} \times 40 \times d^3 = 7.86\, d^3$$

$\therefore$ $d^3 = 215 \times 10^3 / 7.86 = 27.4 \times 10^3$ or $d = 30.1$ say 35 mm **Ans.**

We know that outer diameter of the hub,

$$D = 2d = 2 \times 35 = 70 \text{ mm } \textbf{Ans.}$$

and length of hub, $L = 1.5\,d = 1.5 \times 35 = 52.5$ mm **Ans.**

Let us now check the induced shear stress for the hub material which is cast iron. Considering the hub as a hollow shaft. We know that the maximum torque transmitted (T_{max}).

$$215 \times 10^3 = \frac{\pi}{16} \times \tau_c \left[\frac{D^4 - d^4}{D}\right] = \frac{\pi}{16} \times \tau_c \left[\frac{(70)^4 - (35)^4}{70}\right] = 63\,147\,\tau_c$$

$\therefore$ $\tau_c = 215 \times 10^3 / 63\,147 = 3.4$ N/mm^2 = 3.4 MPa

Since the induced shear stress for the hub material (*i.e.* cast iron) is less than the permissible value of 8 MPa, therefore the design of hub is safe.

2. *Design for key*

Since the crushing stress for the key material is twice its shear stress (*i.e.* $\sigma_{ck} = 2\tau_k$), therefore a square key may be used. From Table 13.1, we find that for a shaft of 35 mm diameter,

Width of key, $w = 12$ mm **Ans.**

and thickness of key, $t = w = 12$ mm **Ans.**

The length of key (l) is taken equal to the length of hub.

$\therefore$ $l = L = 52.5$ mm **Ans.**

Let us now check the induced stresses in the key by considering it in shearing and crushing.

Considering the key in shearing. We know that the maximum torque transmitted (T_{max}),

$$215 \times 10^3 = l \times w \times \tau_k \times \frac{d}{2} = 52.5 \times 12 \times \tau_k \times \frac{35}{2} = 11\ 025\ \tau_k$$

$$\therefore \qquad \tau_k = 215 \times 10^3/11\ 025 = 19.5\ \text{N/mm}^2 = 19.5\ \text{MPa}$$

Considering the key in crushing. We know that the maximum torque transmitted (T_{max}),

$$215 \times 10^3 = l \times \frac{t}{2} \times \sigma_{ck} \times \frac{d}{2} = 52.5 \times \frac{12}{2} \times \sigma_{ck} \times \frac{35}{2} = 5512.5\ \sigma_{ck}$$

$$\therefore \qquad \sigma_{ck} = 215 \times 10^3/\ 5512.5 = 39\ \text{N/mm}^2 = 39\ \text{MPa}$$

Since the induced shear and crushing stresses in the key are less than the permissible stresses, therefore the design for key is safe.

3. *Design for flange*

The thickness of flange (t_f) is taken as 0.5 d.

$$\therefore \qquad t_f = 0.5\ d = 0.5 \times 35 = 17.5\ \text{mm}\ \textbf{Ans.}$$

Let us now check the induced shearing stress in the flange by considering the flange at the junction of the hub in shear.

We know that the maximum torque transmitted (T_{max}),

$$215 \times 10^3 = \frac{\pi\ D^2}{2} \times \tau_c \times t_f = \frac{\pi (70)^2}{2} \times \tau_c \times 17.5 = 134\ 713\ \tau_c$$

$$\therefore \qquad \tau_c = 215 \times 10^3/134\ 713 = 1.6\ \text{N/mm}^2 = 1.6\ \text{MPa}$$

Since the induced shear stress in the flange is less than 8 MPa, therefore the design of flange is safe.

4. *Design for bolts*

Let $\qquad d_1$ = Nominal diameter of bolts.

Since the diameter of the shaft is 35 mm, therefore let us take the number of bolts,

$$n = 3$$

and pitch circle diameter of bolts,

$$D_1 = 3d = 3 \times 35 = 105\ \text{mm}$$

The bolts are subjected to shear stress due to the torque transmitted. We know that the maximum torque transmitted (T_{max}),

$$215 \times 10^3 = \frac{\pi}{4} (d_1)^2\ \tau_b \times n \times \frac{D_1}{2} = \frac{\pi}{4} (d_1)^2\ 40 \times 3 \times \frac{105}{2} = 4950\ (d_1)^2$$

$$\therefore \qquad (d_1)^2 = 215 \times 10^3/4950 = 43.43 \text{ or } d_1 = 6.6\ \text{mm}$$

Assuming coarse threads, the nearest standard size of bolt is M 8. **Ans.**

Other proportions of the flange are taken as follows :

Outer diameter of the flange,

$$D_2 = 4\ d = 4 \times 35 = 140\ \text{mm}\ \textbf{Ans.}$$

Thickness of the protective circumferential flange,

$$t_p = 0.25\ d = 0.25 \times 35 = 8.75 \text{ say } 10\ \text{mm}\ \textbf{Ans.}$$

16. A single plate friction clutch of both sides effective has 300 mm outer diameter and 160 mm inner diameter. The coefficient of friction 0.2 and it runs at 1000 rpm. Find the power transmitted for uniform wear and uniform pressure distributions cases if allowable maximum pressure is 0.08 Mpa.

Given:

$N^1 = I = 2,\ D_0 = 300\text{ mm}\ D_1 = 160\text{mm}\ \mu = 0.2$

$N = 1000\text{ rpm}\ \ p = 0.08\text{ Mpa} = 0.08\text{ N /mm}^2$

Solution:

i. Uniform wear theory

Mean Diameter $D_{m=}\ \dfrac{D_0 - D_1}{2} = \dfrac{300+160}{2} = 230\,mm$

Axial Force

$$\text{Fa} = \frac{1}{2}\pi\ b\ D_1\,(D_0 - D_1)$$

$$\therefore\ \text{Fa} = \frac{1}{2}\pi \times 0.08\ \times 160\ (300 - 160)$$

$$= 2814.87\text{ N}$$

Torque transmitted

$$\text{T} = \frac{1}{2}\ \mu\ n^1\ \ Fa\,Dm$$

$$\text{T} = \frac{1}{2}0.2 \times\ 2\ \times 2814.87 \times 230$$

$$\text{T} = 129484\ \ \text{N} - \text{mm}$$

Power transmitted

$$\text{P} = \frac{2\pi\ M\,T}{60 \times 10^6}$$

$$\text{P} = \frac{2\pi \times 1000 \times 129484}{60 \times 10^6}$$

$$\text{P} = 13.56\ \ \text{kW}$$

ii. Uniform wear theory

Mean Diameter $D_{m=}\ \dfrac{2}{3} = \left(\dfrac{D_0^3 - D_1^3}{D_0^2 - D_1^2}\right)$

$$D_{m=}\ \frac{2}{3} = \left(\frac{300^3 - 160^3}{300^2 - 160^2}\right)$$

$$D_m = 237.1\text{ mm}$$

Axial Force $\quad \text{Fa} = \dfrac{\pi\ \ p\ \ (D_0^2 - D_1^2)}{}$

$$\text{Fa} = \frac{\pi\ 0.08\ \ (300^2 - 160^2)}{4}$$

$$\text{Fa} = 4046.4\text{ N}$$

Torque transmitted

$$\text{T} = n^1\ \frac{1}{2}\ \mu\ \ \ Fa\,Dm$$

$$T = 2\ \frac{1}{2} 0.2 \times 4046.4 \times 237.1$$

$$T = 191880.3 \ \text{N} - \text{mm}$$

Power transmitted

$$P = \frac{2\pi \ nT}{60 \times 10^6}$$

$$P = \frac{2 \ \times \pi \times 1000 \times 191880.3}{60 \times 10^6}$$

$$P = 20.1 \ \text{kW}$$

Unit IV

17. Design a knuckle joint to transmit 150 kN. The design stresses may be taken as 75 MPa in tension, 60 MPa in shear and 150 MPa in compression.

Given: P = 150 kN = 150×10^3 N; σ_t = 75 MPa = 75 N/mm^2; τ = 60 MPa = 60 N/mm2; σ_c = 150 MPa = 150 N/mm^2

The joint is designed by considering the various methods of failure as discussed below:

1. Failure of the solid rod in tension

Let d = Diameter of the rod.

We know that the load transmitted (P),

$$150 \times 10^3 = \frac{\pi}{4} \times d^2 \times \sigma_t = \frac{\pi}{4} \times d^2 \times 75 = 59\, d^2$$

$\therefore$ $d^2 = 150 \times 10^3 / 59 = 2540$ or $d = 50.4$ say 52 mm **Ans.**

Now the various dimensions are fixed as follows :

Diameter of knuckle pin,

$$d_1 = d = 52 \text{ mm}$$

Outer diameter of eye, $d_2 = 2\, d = 2 \times 52 = 104$ mm

Diameter of knuckle pin head and collar,

$$d_3 = 1.5\, d = 1.5 \times 52 = 78 \text{ mm}$$

Thickness of single eye or rod end,

$$t = 1.25\, d = 1.25 \times 52 = 65 \text{ mm}$$

Thickness of fork, $t_1 = 0.75\, d = 0.75 \times 52 = 39$ say 40 mm

Thickness of pin head, $t_2 = 0.5\, d = 0.5 \times 52 = 26$ mm

2. Failure of the knuckle pin in shear

Since the knuckle pin is in double shear, therefore load (P),

$$150 \times 10^3 = 2 \times \frac{\pi}{4} \times (d_1)^2\, \tau = 2 \times \frac{\pi}{4} \times (52)^2\, \tau = 4248\, \tau$$

$\therefore$ $\tau = 150 \times 10^3 / 4248 = 35.3 \text{ N/mm}^2 = 35.3 \text{ MPa}$

3. Failure of the single eye or rod end in tension

The single eye or rod end may fail in tension due to the load. We know that load (P),

$$150 \times 10^3 = (d_2 - d_1)\, t \times \sigma_t = (104 - 52)\, 65 \times \sigma_t = 3380\, \sigma_t$$

$\therefore$ $\sigma_t = 150 \times 10^3 / 3380 = 44.4 \text{ N/mm}^2 = 44.4 \text{ MPa}$

4. *Failure of the single eye or rod end in shearing*

The single eye or rod end may fail in shearing due to the load. We know that load (P),

$$150 \times 10^3 = (d_2 - d_1)\, t \times \tau = (104 - 52)\, 65 \times \tau = 3380\, \tau$$

$$\therefore \quad \tau = 150 \times 10^3 / 3380 = 44.4 \text{ N/mm}^2 = 44.4 \text{ MPa}$$

5. *Failure of the single eye or rod end in crushing*

The single eye or rod end may fail in crushing due to the load. We know that load (P),

$$150 \times 10^3 = d_1 \times t \times \sigma_c = 52 \times 65 \times \sigma_c = 3380\, \sigma_c$$

$$\therefore \quad \sigma_c = 150 \times 10^3 / 3380 = 44.4 \text{ N/mm}^2 = 44.4 \text{ MPa}$$

6. *Failure of the forked end in tension*

The forked end may fail in tension due to the load. We know that load (P),

$$150 \times 10^3 = (d_2 - d_1)\, 2\, t_1 \times \sigma_t = (104 - 52)\, 2 \times 40 \times \sigma_t = 4160\, \sigma_t$$

$$\therefore \quad \sigma_t = 150 \times 10^3 / 4160 = 36 \text{ N/mm}^2 = 36 \text{ MPa}$$

7. *Failure of the forked end in shear*

The forked end may fail in shearing due to the load. We know that load (P),

$$150 \times 10^3 = (d_2 - d_1)\, 2\, t_1 \times \tau = (104 - 52)\, 2 \times 40 \times \tau = 4160\, \tau$$

$$\therefore \quad \tau = 150 \times 10^3 / 4160 = 36 \text{ N/mm}^2 = 36 \text{ MPa}$$

8. *Failure of the forked end in crushing*

The forked end may fail in crushing due to the load. We know that load (P),

$$150 \times 10^3 = d_1 \times 2\, t_1 \times \sigma_c = 52 \times 2 \times 40 \times \sigma_c = 4160\, \sigma_c$$

$$\therefore \quad \sigma_c = 150 \times 10^3 / 4180 = 36 \text{ N/mm}^2 = 36 \text{ MPa}$$

From above, we see that the induced stresses are less than the given design stresses, therefore the joint is safe.

18. A helical compression spring made of oil tempered carbon steel, is subjected to a load which varies from 400 N to 1000 N. The spring index is 6 and the design factor of safety is 1.25. If the yield stress in shear is 770 MPa and endurance stress in shear is 350 MPa, find: 1. Size of the spring wire, 2. Diameters of the spring, 3. Number of turns of the spring, and 4. Free length of the spring. The compression of the spring at the maximum load is 30 mm. The modulus of rigidity for the spring material may be taken as 80 kN/mm^2.

Solution. Given : W_{min} = 400 N ; W_{max} = 1000 N ; C = 6 ; $F.S.$ = 1.25 ; τ_y = 770 MPa = 770 N/mm^2 ; τ_e = 350 MPa = 350 N/mm^2 ; δ = 30 mm ; G = 80 kN/mm^2 = 80 × 10^3 N/mm^2

1. *Size of the spring wire*

Let $\quad d$ = Diameter of the spring wire, and

D = Mean diameter of the spring = $C.d$ = 6 d $\quad$ ... ($\because D/d = C = 6$)

We know that the mean load,

$$W_m = \frac{W_{max} + W_{min}}{2} = \frac{1000 + 400}{2} = 700 \text{ N}$$

and variable load, $$W_v = \frac{W_{max} - W_{min}}{2} = \frac{1000 - 400}{2} = 300 \text{ N}$$

Shear stress factor,

$$K_S = 1 + \frac{1}{2C} = 1 + \frac{1}{2 \times 6} = 1.083$$

Wahl's stress factor,

$$K = \frac{4C-1}{4C-4} + \frac{0.615}{C} = \frac{4 \times 6 - 1}{4 \times 6 - 4} + \frac{0.615}{6} = 1.2525$$

We know that mean shear stress,

$$\tau_m = K_S \times \frac{8\, W_m \times D}{\pi\, d^3} = 1.083 \times \frac{8 \times 700 \times 6\, d}{\pi\, d^3} = \frac{11\,582}{d^2} \text{ N/mm}^2$$

and variable shear stress,

$$\tau_v = K \times \frac{8\, W_v \times D}{\pi\, d^3} = 1.2525 \times \frac{8 \times 300 \times 6\, d}{\pi\, d^3} = \frac{5740}{d^2} \text{ N/mm}^2$$

We know that

$$\frac{1}{F.S.} = \frac{\tau_m - \tau_v}{\tau_y} + \frac{2\,\tau_v}{\tau_e}$$

$$\frac{1}{1.25} = \frac{\frac{11\,582}{d^2} - \frac{5740}{d^2}}{770} + \frac{2 \times \frac{5740}{d^2}}{350} = \frac{7.6}{d^2} + \frac{32.8}{d^2} = \frac{40.4}{d^2}$$

$\therefore$ $d^2 = 1.25 \times 40.4 = 50.5$ or $d = 7.1$ mm **Ans.**

2. Diameters of the spring

We know that mean diameter of the spring,

$$D = C.d = 6 \times 7.1 = 42.6 \text{ mm } \textbf{Ans.}$$

Outer diameter of the spring,

$$D_o = D + d = 42.6 + 7.1 = 49.7 \text{ mm } \textbf{Ans.}$$

and inner diameter of the spring,

$$D_i = D - d = 42.6 - 7.1 = 35.5 \text{ mm } \textbf{Ans.}$$

3. Number of turns of the spring

Let n = Number of active turns of the spring.

We know that deflection of the spring (δ),

$$30 = \frac{8\, W.D^3.n}{G.d^4} = \frac{8 \times 1000\,(42.6)^3\, n}{80 \times 10^3\,(7.1)^4} = 3.04\, n$$

$\therefore$ $n = 30 / 3.04 = 9.87$ say 10 **Ans.**

Assuming the ends of the spring to be squared and ground, the total number of turns of the spring,

$$n' = n + 2 = 10 + 2 = 12 \textbf{ Ans.}$$

4. Free length of the spring

We know that free length of the spring,

$$L_F = n'.d + \delta + 0.15\,\delta = 12 \times 7.1 + 30 + 0.15 \times 30 \text{ mm}$$

$$= 119.7 \text{ say } 120 \text{ mm } \textbf{Ans.}$$

Unit V

19. A shaft is supported by two bearings placed 1 m apart. A 600 mm diameter pulley is mounted at a distance of 300 mm to the right of left hand bearing and this drives a pulley directly below it with the help of belt having maximum tension of 2.25 kN. Another pulley 400 mm diameter is placed 200 mm to the left of right hand bearing and is driven with the help of electric motor and belt, which is placed horizontally to the right. The angle of contact for both the pulleys is 180° and μ = 0.24. Determine the suitable diameter for a solid

shaft, allowing working stress of 63 MPa in tension and 42 MPa in shear for the material of shaft. Assume that the torque on one pulley is equal to that on the other pulley.

Solution. Given : $AB = 1$ m ; $D_C = 600$ mm or $R_C = 300$ mm $= 0.3$ m ; $AC = 300$ mm $= 0.3$ m ; $T_1 = 2.25$ kN $= 2250$ N ; $D_D = 400$ mm or $R_D = 200$ mm $= 0.2$ m ; $BD = 200$ mm $= 0.2$ m ; $\theta = 180° = \pi$ rad ; $\mu = 0.24$; $\sigma_b = 63$ MPa $= 63$ N/mm^2 ; $\tau = 42$ MPa $= 42$ N/mm^2

The space diagram of the shaft is shown in Fig. 14.5 (*a*).

Let T_1 = Tension in the tight side of the belt on pulley $C = 2250$ N ...(Given)

T_2 = Tension in the slack side of the belt on pulley C.

We know that

$$2.3 \log\left(\frac{T_1}{T_2}\right) = \mu.\theta = 0.24 \times \pi = 0.754$$

$$\therefore \quad \log\left(\frac{T_1}{T_2}\right) = \frac{0.754}{2.3} = 0.3278 \quad \text{or} \quad \frac{T_1}{T_2} = 2.127 \qquad \text{...(Taking antilog of 0.3278)}$$

and

$$T_2 = \frac{T_1}{2.127} = \frac{2250}{2.127} = 1058 \text{ N}$$

$\therefore$ Vertical load acting on the shaft at C,

$$W_C = T_1 + T_2 = 2250 + 1058 = 3308 \text{ N}$$

and vertical load on the shaft at D

$$= 0$$

The vertical load diagram is shown in Fig. 14.5 (*c*).

We know that torque acting on the pulley C,

$$T = (T_1 - T_2)\, R_C = (2250 - 1058)\, 0.3 = 357.6 \text{ N-m}$$

The torque diagram is shown in Fig. 14.5 (*b*).

Let T_3 = Tension in the tight side of the belt on pulley D, and

T_4 = Tension in the slack side of the belt on pulley D.

Since the torque on both the pulleys (*i.e.* C and D) is same, therefore

$$(T_3 - T_4)\, R_D = T = 357.6 \text{ N-m} \quad \text{or} \quad T_3 - T_4 = \frac{357.6}{R_D} = \frac{357.6}{0.2} = 1788 \text{ N} \qquad \text{...}(i)$$

We know that

$$= \frac{T_3}{T_4} = \frac{T_1}{T_2} = 2.127 \quad \text{or} \quad T_3 = 2.127\, T_4 \qquad \text{...}(ii)$$

From equations (*i*) and (*ii*), we find that

$$T_3 = 3376 \text{ N, and } T_4 = 1588 \text{ N}$$

$\therefore$ Horizontal load acting on the shaft at D,

$$W_D = T_3 + T_4 = 3376 + 1588 = 4964 \text{ N}$$

and horizontal load on the shaft at $C = 0$

The horizontal load diagram is shown in Fig. 14.5 (*d*).

Now let us find the maximum bending moment for vertical and horizontal loading.

First of all, considering the vertical loading at C. Let R_{AV} and R_{BV} be the reactions at the bearings A and B respectively. We know that

$$R_{AV} + R_{BV} = 3308 \text{ N}$$

Taking moments about A,

$$R_{BV} \times 1 = 3308 \times 0.3 \quad \text{or} \quad R_{BV} = 992.4 \text{ N}$$

and
$$R_{AV} = 3308 - 992.4 = 2315.6 \text{ N}$$

We know that B.M. at A and B,

$$M_{AV} = M_{BV} = 0$$

B.M. at C, $\quad M_{CV} = R_{AV} \times 0.3 = 2315.6 \times 0.3 = 694.7$ N-m

B.M. at D, $\quad M_{DV} = R_{BV} \times 0.2 = 992.4 \times 0.2 = 198.5$ N-m

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Taking moments about A,

$$R_{BV} \times 1 = 3308 \times 0.3 \quad \text{or} \quad R_{BV} = 992.4 \text{ N}$$

and
$$R_{AV} = 3308 - 992.4 = 2315.6 \text{ N}$$

We know that B.M. at A and B,

$$M_{AV} = M_{BV} = 0$$

B.M. at C, $\quad M_{CV} = R_{AV} \times 0.3 = 2315.6 \times 0.3 = 694.7$ N-m

B.M. at D, $\quad M_{DV} = R_{BV} \times 0.2 = 992.4 \times 0.2 = 198.5$ N-m

First of all, considering the vertical loading at C. Let R_{AV} and R_{BV} be the reactions at the bearings A and B respectively. We know that

$$R_{AV} + R_{BV} = 3308 \text{ N}$$

Taking moments about A,

$$R_{BV} \times 1 = 3308 \times 0.3 \quad \text{or} \quad R_{BV} = 992.4 \text{ N}$$

and
$$R_{AV} = 3308 - 992.4 = 2315.6 \text{ N}$$

We know that B.M. at A and B,

$$M_{AV} = M_{BV} = 0$$

B.M. at C, $\quad M_{CV} = R_{AV} \times 0.3 = 2315.6 \times 0.3 = 694.7$ N-m

B.M. at D, $\quad M_{DV} = R_{BV} \times 0.2 = 992.4 \times 0.2 = 198.5$ N-m

The bending moment diagram for vertical loading in shown in Fig. 14.5 (*e*).

Now considering horizontal loading at D. Let R_{AH} and R_{BH} be the reactions at the bearings A and B respectively. We know that

$$R_{AH} + R_{BH} = 4964 \text{ N}$$

Taking moments about A,

$$R_{BH} \times 1 = 4964 \times 0.8 \quad \text{or} \quad R_{BH} = 3971 \text{ N}$$

and
$$R_{AH} = 4964 - 3971 = 993 \text{ N}$$

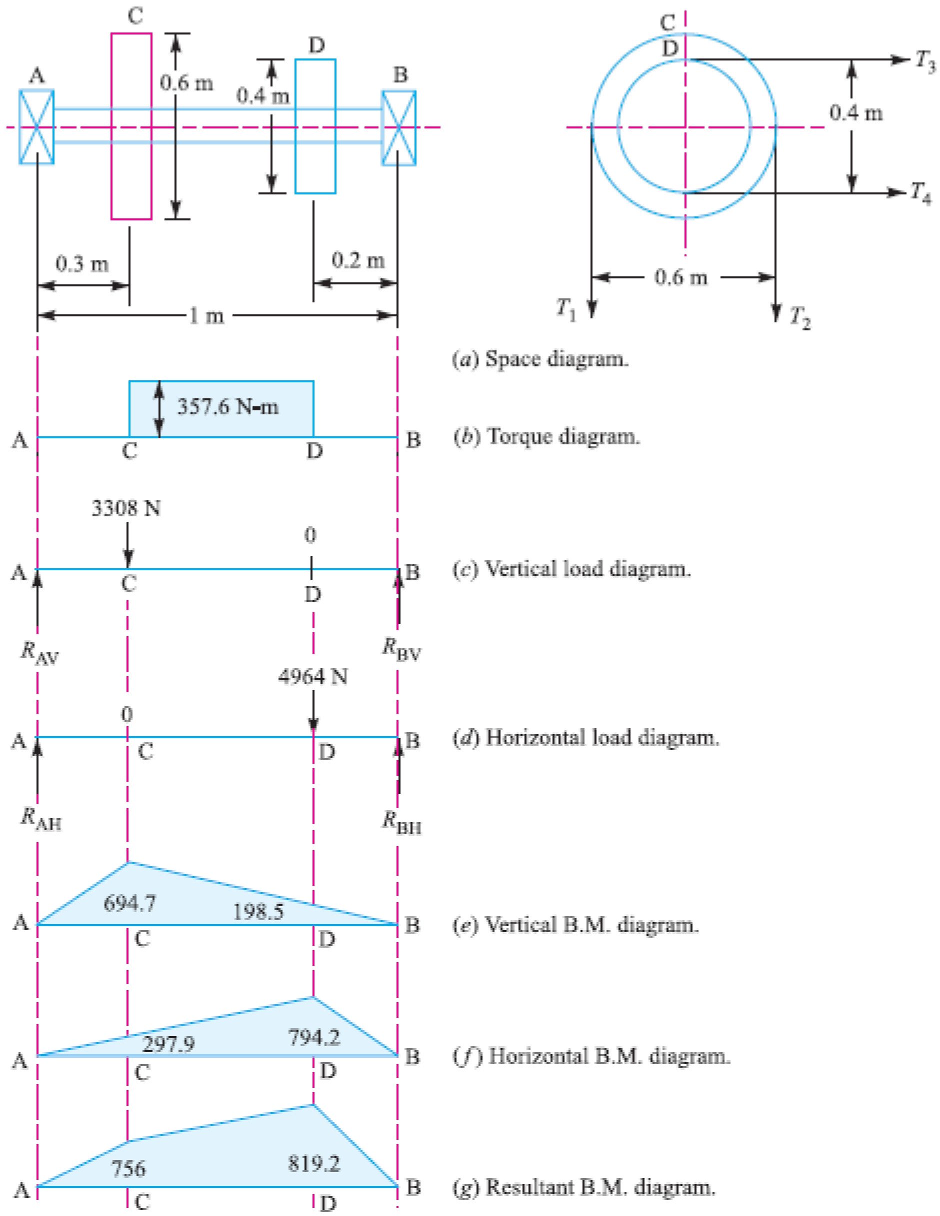
We know that B.M. at A and B,

$$M_{AH} = M_{BH} = 0$$

B.M. at C, $\quad M_{CH} = R_{AH} \times 0.3 = 993 \times 0.3 = 297.9$ N-m

B.M. at D, $\quad M_{DH} = R_{BH} \times 0.2 = 3971 \times 0.2 = 794.2$ N-m

The bending moment diagram for horizontal loading is shown in Fig. 14.5 (*f*).



(*a*) Space diagram.

(*b*) Torque diagram.

(*c*) Vertical load diagram.

(*d*) Horizontal load diagram.

(*e*) Vertical B.M. diagram.

(*f*) Horizontal B.M. diagram.

(*g*) Resultant B.M. diagram.

Resultant B.M. at *C*,

$$M_C = \sqrt{(M_{CV})^2 + (M_{CH})^2} = \sqrt{(694.7)^2 + (297.9)^2} = 756 \text{ N-m}$$

and resultant B.M. at *D*,

$$M_D = \sqrt{(M_{DV})^2 + (M_{DH})^2} = \sqrt{(198.5)^2 + (794.2)^2} = 819.2 \text{ N-m}$$

The resultant bending moment diagram is shown in Fig. 14.5 (*g*).

We see that bending moment is maximum at *D*.

$\therefore$ Maximum bending moment,

$$M = M_D = 819.2 \text{ N-m}$$

Let d = Diameter of the shaft.

We know that equivalent twisting moment,

$$T_e = \sqrt{M^2 + T^2} = \sqrt{(819.2)^2 + (357.6)^2} = 894 \text{ N-m}$$
$$= 894 \times 10^3 \text{ N-mm}$$

We also know that equivalent twisting moment (T_e),

$$894 \times 10^3 = \frac{\pi}{16} \times \tau \times d^3 = \frac{\pi}{16} \times 42 \times d^3 = 8.25\, d^3$$

$\therefore$ $d^3 = 894 \times 10^3 / 8.25 = 108 \times 10^3$ or $d = 47.6$ mm

Again we know that equivalent bending moment,

$$M_e = \frac{1}{2}\left(M + \sqrt{M^2 + T^2}\right) = \frac{1}{2}(M + T_e)$$

$$= \frac{1}{2}(819.2 + 894) = 856.6 \text{ N-m} = 856.6 \times 10^3 \text{ N-mm}$$

We also know that equivalent bending moment (M_e),

$$856.6 \times 10^3 = \frac{\pi}{32} \times \sigma_b \times d^3 = \frac{\pi}{32} \times 63 \times d^3 = 6.2\, d^3$$

$\therefore$ $d^3 = 856.6 \times 10^3 / 6.2 = 138.2 \times 10^3$ or $d = 51.7$ mm

Taking larger of the two values, we have

$d = 51.7$ say 55 mm **Ans.**

20. Determine the diameter of a circular rod made of ductile material with a fatigue strength (complete stress reversal), σ_{-1} = 265 MPa and a tensile yield strength of 350 MPa. The member is subjected to a varying axial load from $W_{min} = -300 \times 10^3$ N to $W_{max} = 700 \times 10^3$ N and has a stress concentration factor = 1.8. Use factor of safety as 2.0.

Solution. Given : $\sigma_{-1} = 265$ MPa $= 265$ N/mm^2 ; $\sigma_y = 350$ MPa $= 350$ N/mm^2 ; $W_{min} = -300 \times 10^3$ N ; $W_{max} = 700 \times 10^3$ N ; $K_f = 1.8$; $F.S. = 2$

Let d = Diameter of the circular rod in mm.

$\therefore$ Area, $A = \frac{\pi}{4} \times d^2 = 0.7854\, d^2$ mm^2

We know that the mean or average load,

$$W_m = \frac{W_{max} + W_{min}}{2} = \frac{700 \times 10^3 + (-300 \times 10^3)}{2} = 200 \times 10^3 \text{ N}$$

$\therefore$ Mean stress, $\sigma_m = \frac{W_m}{A} = \frac{200 \times 10^3}{0.7854\, d^2} = \frac{254.6 \times 10^3}{d^2}$ N/mm^2

Variable load, $W_a = \frac{W_{max} - W_{min}}{2} = \frac{700 \times 10^3 - (-300 \times 10^3)}{2} = 500 \times 10^3$ N

$\therefore$ Variable stress, $\sigma_a = \frac{W_a}{A} = \frac{500 \times 10^3}{0.7854\, d^2} = \frac{636.5 \times 10^3}{d^2}$ N/mm^2

We know that according to Soderberg's formula,

$$\frac{1}{F.S.} = \frac{\sigma_m}{\sigma_y} + \frac{\sigma_a \times K_f}{\sigma_{-1}}$$

$$\frac{1}{2} = \frac{254.6 \times 10^3}{d^2 \times 350} + \frac{636.5 \times 10^3 \times 1.8}{d^2 \times 265} = \frac{727}{d^2} + \frac{4323}{d^2} = \frac{5050}{d^2}$$

$\therefore$ $d^2 = 5050 \times 2 = 10\,100$ or $d = 100.5$ mm **Ans.**

DESIGN OF MACHINE ELEMENTS

Question Paper Code – 4615107

Part A

1. **What are the various theories of failure?**

The various theories of failure are

- ✓ Maximum principal (or normal) stress theory (also known as Rankine's theory).
- ✓ Maximum shear stress theory (also known as Guest's or Tresca's theory).
- ✓ Maximum principal (or normal) strain theory (also known as Saint Venant theory).
- ✓ Maximum strain energy theory (also known as Haigh's theory).
- ✓ Maximum distortion energy theory (also known as Hencky and Von Mises theory).

2. **Define but and lap joint?**

- ✓ The **butt joint** is obtained by placing the plates edge to edge welding the edges of the plates
- ✓ The **lap joint** or the fillet joint is obtained by overlapping the plates and then welding the edges of the plates.

3. **What is the meaning of bolt M24 x2?**

Where,

M - Metric standard

24 - Diameter of bolt

P – Pitch of the thread,

4. **What is Stud?**

- ✓ A stud is a round bar threaded at both ends. One end of the stud is screwed into a tapped hole of the parts to be fastened, while the other end receives a nut on it.

5. **What is the function of a coupling between two shafts?**

- ✓ To provide for the connection of shafts of units that are manufactured separately such as a motor and generator and to provide for disconnection for repairs or alternations.
- ✓ To provide for misalignment of the shafts or to introduce mechanical flexibility.
- ✓ To reduce the transmission of shock loads from one shaft to another.
- ✓ To introduce protection against overloads.

6. **Differentiate between Self-energizing and self locking brakes.**

- ✓ The frictional force helps to apply the brake. Such type of brakes are said to be **self energizing brakes.**

- ✓ When the frictional force is great enough to apply the brake with no external force, then the brake is said to be **self-locking brake.**

7. **What is spring Index?**

- ✓ The spring index is defined as the ratio of the mean diameter of the coil to the diameter of the wire. Mathematically,

Spring index, $C = D / d$

Where

D = Mean diameter of the coil, and

d = Diameter of the wire.

8. **Name some application of cotter joint.**

- ✓ Cotter joint is usually used in
  - Connecting a piston rod to the crosshead of a reciprocating steam engine,
  - A piston rod and its extension as a tail or pump rod,
  - Strap end of connecting rod

9. **What is simple torsion?**

- ✓ When a machine member is subjected to the action of two equal and opposite couples acting in parallel planes (or torque or twisting moment), then the machine member is said to be subjected to torsion.

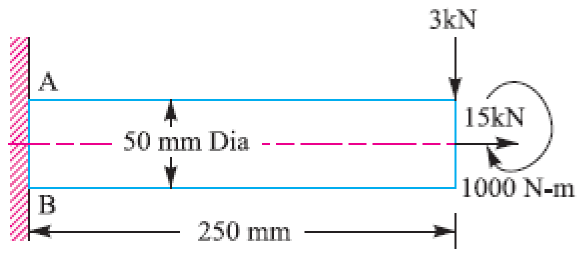
10. **What for Goodman and Soderberg diagram are used?**

- ✓ A Goodman and Soderberg diagram is a graph of (linear) mean stress against (linear) alternating stress, showing when the material fails at some given number of cycles.

Part B

11. **A cantilever of span 250mm and 50mm diameter carries a vertical downward load of 3kN at free end along with a torque of 1000N-m and an axial pull of 15kN.Calculate the maximum normal stress at the top and bottom face of the fixed end.**

Given: W = 3 kN = 3000 N; T = 1000 N-m = 1 × 106 N-mm; P = 15 kN = 15 × 103 N; d = 50 mm; x = 250 mm



We know that cross-sectional area of the shaft

$$A = \frac{\pi}{4} \times d^2$$

$$= \frac{\pi}{4} (50)^2 = 1964 \text{ mm}^2$$

∴ Tensile stress due to axial pulling at points *A* and *B*,

$$\sigma_o = \frac{P}{A} = \frac{15 \times 10^3}{1964} = 7.64 \text{ N/mm}^2 = 7.64 \text{ MPa}$$

Bending moment at points *A* and *B*,

$$M = W.x = 3000 \times 250 = 750 \times 10^3 \text{ N-mm}$$

Section modulus for the shaft,

$$Z = \frac{\pi}{32} \times d^3 = \frac{\pi}{32} (50)^3$$

$$= 12.27 \times 10^3 \text{ mm}^3$$

∴ Bending stress at points *A* and *B*,

$$\sigma_b = \frac{M}{Z} = \frac{750 \times 10^3}{12.27 \times 10^3}$$

$$= 61.1 \text{ N/mm}^2 = 61.1 \text{ MPa}$$

This bending stress is tensile at point *A* and compressive at point *B*.

∴ Resultant tensile stress at point *A*,

$$\sigma_A = \sigma_b + \sigma_o = 61.1 + 7.64$$
$$= 68.74 \text{ MPa}$$

and resultant compressive stress at point *B*,

$$\sigma_B = \sigma_b - \sigma_o = 61.1 - 7.64 = 53.46 \text{ MPa}$$

We know that the shear stress at points *A* and *B* due to the torque transmitted,

$$\tau = \frac{16\ T}{\pi\ d^3} = \frac{16 \times 1 \times 10^6}{\pi\ (50)^3} = 40.74 \text{ N/mm}^2 = 40.74 \text{ MPa} \qquad \ldots \left(\because\ T = \frac{\pi}{16} \times \tau \times d^3 \right)$$

Stresses at point A

We know that maximum principal (or normal) stress at point *A*,

$$\sigma_{A(max)} = \frac{\sigma_A}{2} + \frac{1}{2}\left[\sqrt{(\sigma_A)^2 + 4\ \tau^2}\right]$$

$$= \frac{68.74}{2} + \frac{1}{2}\left[\sqrt{(68.74)^2 + 4\ (40.74)^2}\right]$$

$$= 34.37 + 53.3 = 87.67 \text{ MPa (tensile)}$$

Stresses at point B

We know that maximum principal (or normal) stress at point *B*,

$$\sigma_{B(max)} = \frac{\sigma_B}{2} + \frac{1}{2}\left[\sqrt{(\sigma_B)^2 + 4\,\tau^2}\right]$$

$$= \frac{53.46}{2} + \frac{1}{2}\left[\sqrt{(53.46)^2 + 4\,(40.74)^2}\right]$$

$$= 26.73 + 48.73 = 75.46 \text{ MPa (compressive)}$$

12. A plate 100 mm wide and 12.5 mm thick is to be welded to another plate by means of parallel fillet welds. The plates are subjected to a load of 50 kN. Find the length of the weld so that the maximum stress does not exceed 56 MPa. Consider the joint first under static loading and then under fatigue loading.

Given: *Width = 100 mm ; Thickness = 12.5 mm ; P = 50 kN = 50 × 10^3N ; τ = 56 MPa = 56 N/mm^2

Length of weld for static loading

Let l = Length of weld, and

s = Size of weld = Plate thickness

= 12.5 mm ... (Given)

We know that the maximum load which the plates can carry for double parallel fillet welds (P),

50 × 103 =1.414 $s \times l \times \tau$

= 1.414 × 12.5 × l × 56

= 990 l

∴ l = 50 × 103 / 990 = 50.5 mm

Adding 12.5 mm for starting and stopping of weld run, we have

l = 50.5 + 12.5 = 63 mm **Ans.**

Length of weld for fatigue loading

From the data book we find that the stress concentration factor for parallel fillet welding is 2.7.

∴ Permissible shear stress,

τ = 56 / 2.7 = 20.74 N/mm^2

We know that the maximum load which the plates can carry for double parallel fillet welds (P),

50 × 103 = 1.414 $s \times l \times \tau$

= 1.414 × 12.5 × l × 20.74

= 367 l

∴ l = 50 × 103 / 367 = 136.2 mm

Adding 12.5 for starting and stopping of weld run, we have

l = 136.2 + 12.5 = 148.7 mm **Ans.**

13. A cast iron cylinder head is fastened to a cylinder of 500 mm bore with 8 stud bolts. The maximum pressure inside the cylinder is 2MPa. Stiffness of the part is thrice the stiffness

of the bolt. What should be the initial tightening load so that the joint is leak proof at maximum pressure? Also choose a suitable bolt for the above application.

Given: D= 500 mm; n= 8; Max. Pressure p= 2MPa; $k_p=3k_b$;

Solution:

We know that load acting on cylinder head,

F = Area x pressure = – x p

$$= \frac{\pi}{4}(500)^2 \times 2$$

= 392699.08 N

Therefore load on each bolt is

F_o = load/ No of bolt

=392699.08/ 8

= 49087.385 N

This will causes the opening of the joint in case the initial load F_i is not sufficient.

We know that

(——)

(–)

= 36065.54 N

Thus, a slightly greater load will ensure a leak proof joint; hence the load of 37000 N may be used.

For size of the bolt,

Load on the bolt (P) = 49087 N

Assume tensile stress of the bolt = 50 N/mm²

Core diameter of the threaded part (d_c) =?

–

–

d_c^2 = 1249.99

d_c = 35.35

From the data book Page: No.:5.42 we find that standard core diameter is 36.402 mm and the corresponding size of the bolt is M36 x 4.

14. Determine the size of the bolts and the thickness of the arm for the bracket as shown in Fig., if it carries a load of 40 kN at an angle of 60° to the vertical. The material of the

bracket and the bolts is same for which the safe stresses can be assumed as70, 50 and 150 MPa in tension, shear and compression respectively.

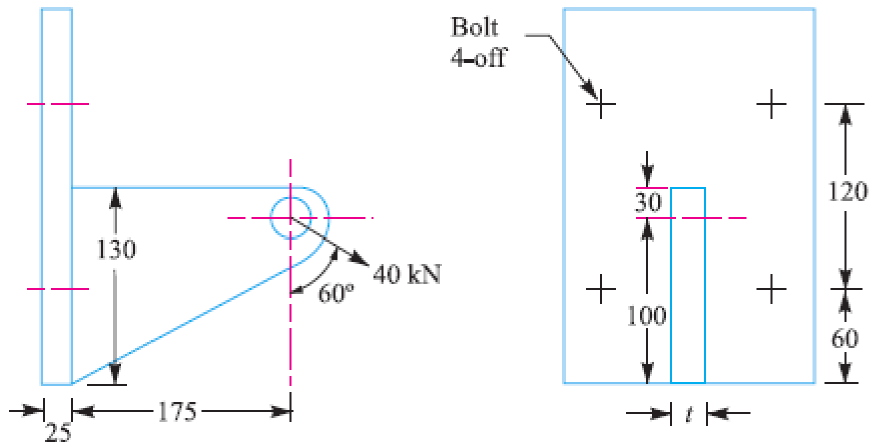
Solution. Given : $W = 40$ kN $= 40 \times 10^3$ N ; $\sigma_t = 70$ MPa $= 70$N/mm^2 ; $\tau = 50$ MPa $= 50$ N/mm^2; $\sigma_c = 150$ MPa $= 150$ N/mm^2

Since the load $W = 40$ kN is inclined at an angle of 60° to the vertical, therefore resolving it into horizontal and vertical components. We know that horizontal component of 40 kN,

$$W_H = 40 \times \sin 60° = 40 \times 0.866 = 34.64 \text{ kN} = 34\,640 \text{ N}$$

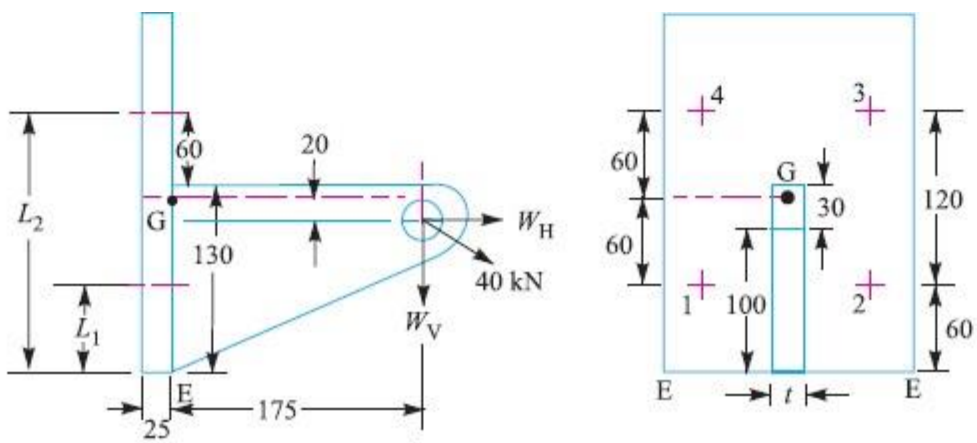
and vertical component of 40 kN,

$$W_V = 40 \times \cos 60° = 40 \times 0.5 = 20 \text{ kN} = 20\,000 \text{ N}$$



All dimensions in mm.

Due to the horizontal component (W_H), which acts parallel to the axis of the bolts as shown in Fig the following two effects are produced :



All dimensions in mm.

1. A direct tensile load equally shared by all the four bolts, and
2. A turning moment about the centre of gravity of the bolts, in the anticlockwise direction.

∴ Direct tensile load on each bolt,

$$W_{t1} = \frac{W_H}{4} = \frac{34\,640}{4} = 8660 \text{ N}$$

Since the centre of gravity of all the four bolts lies in the centre at G (because of symmetrical bolts), therefore the turning moment is in the anticlockwise direction. From the geometry of the Fig. . we find that the distance of horizontal component from the centre of gravity (G) of the bolts

$= 60 + 60 - 100 = 20$ mm

$\therefore$ Turning moment due to W_H about G,

$$T_H = W_H \times 20 = 34\ 640 \times 20 = 692.8 \times 10^3 \text{ N-mm} \quad \text{...(Anticlockwise)}$$

Due to the vertical component W_V, which acts perpendicular to the axis of the bolts as shown in Fig. the following two effects are produced:

1. A direct shear load equally shared by all the four bolts, and
2. A turning moment about the edge of the bracket in the clockwise direction.

$\therefore$ Direct shear load on each bolt,

$$W_s = \frac{W_V}{4} = \frac{20\ 000}{4} = 5000 \text{ N}$$

Distance of vertical component from the edge E of the bracket,

$= 175$ mm

$\therefore$ Turning moment due to W_V about the edge of the bracket,

$$T_V = W_V \times 175 = 20\ 000 \times 175 = 3500 \times 10^3 \text{ N-mm} \quad \text{(Clockwise)}$$

From above, we see that the clockwise moment is greater than the anticlockwise moment, therefore,

$$\text{Net turning moment} = 3500 \times 10^3 - 692.8 \times 10^3 = 2807.2 \times 10^3 \text{ N-mm (Clockwise)} \quad \text{...}(i)$$

Due to this clockwise moment, the bracket tends to tilt about the lower edge E.

Let w = Load on each bolt per mm distance from the edge E due to the turning effect of the bracket,

L_1 = Distance of bolts 1 and 2 from the tilting edge E = 60 mm, and

L_2 = Distance of bolts 3 and 4 from the tilting edge E

$= 60 + 120 = 180$ mm

$\therefore$ Total moment of the load on the bolts about the tilting edge E

$$= 2\,(w.L_1)\,L_1 + 2\,(w.L_2)\,L_2$$

... ($\because$ There are two bolts each at distance L_1 and L_2.)

$$= 2w\,(L_1)^2 + 2w(L_2)^2 = 2w\,(60)^2 + 2w(180)^2$$

$$= 72\ 000\ w \text{ N-mm} \quad \text{...}(ii)$$

From equations (*i*) and (*ii*),

$$w = 2807.2 \times 10^3 / 72\ 000 = 39 \text{ N/mm}$$

Since the heavily loaded bolts are those which lie at a greater distance from the tilting edge, therefore the upper bolts 3 and 4 will be heavily loaded. Thus the diameter of the bolt should be based on the load on the upper bolts. We know that the maximum tensile load on each upper bolt,

$$W_{t2} = w.L_2 = 39 \times 180 = 7020 \text{ N}$$

$\therefore$ Total tensile load on each of the upper bolt,

$$W_t = W_{t1} + W_{t2} = 8660 + 7020 = 15\ 680 \text{ N}$$

Since each upper bolt is subjected to a tensile load (W_t = 15 680 N) and a shear load (W_s = 5000 N), therefore equivalent tensile load,

$$W_{te} = \frac{1}{2}\left[W_t + \sqrt{(W_t)^2 + 4(W_s)^2}\right]$$

$$= \frac{1}{2}\left[15\ 680 + \sqrt{(15\ 680)^2 + 4(5000)^2}\right] \text{ N}$$

$$= \frac{1}{2}[15\ 680 + 18\ 600] = 17\ 140 \text{ N} \qquad ...(iii)$$

Size of the bolts

Let d_c = Core diameter of the bolts.

We know that tensile load on each bolt

$$= \frac{\pi}{2}(d_c)^2 \sigma_t = \frac{\pi}{4}(d_c)^2\ 70 = 55\ (d_c)^2 \text{ N} \qquad ...(iv)$$

From equations ***(iii)*** and ***(iv)***, we get

$$(d_c)^2 = 17\ 140 / 55 = 311.64 \quad \text{or} \quad d_c = 17.65 \text{ mm}$$

From Table 11.1 (coarse series), we find that the standard core diameter is 18.933 mm and corresponding size of the bolt is M 22. **Ans.**

Thickness of the arm of the bracket

Let t = Thickness of the arm of the bracket in mm, and

b = Depth of the arm of the bracket = 130 mm ...(Given)

We know that cross-sectional area of the arm,

$$A = b \times t = 130\ t \text{ mm}^2$$

and section modulus of the arm,

$$Z = \frac{1}{6}\,t\,(b)^2 = \frac{1}{6} \times t\,(130)^2 = 2817\ t \text{ mm}^3$$

Due to the horizontal component W_H, the following two stresses are induced in the arm :

1. Direct tensile stress,

$$\sigma_{t1} = \frac{W_H}{A} = \frac{34\ 640}{130\ t} = \frac{266.5}{t} \text{ N/mm}^2$$

2. Bending stress causing tensile in the upper most fibres of the arm and compressive in the lower most fibres of the arm. We know that the bending moment of W_H about the centre of gravity of the arm,

$$M_H = W_H\left(100 - \frac{130}{2}\right) = 34\ 640 \times 35 = 1212.4 \times 10^3 \text{ N-mm}$$

$\therefore$ Bending stress, $$\sigma_{t2} = \frac{M_H}{Z} = \frac{1212.4 \times 10^3}{2817\ t} = \frac{430{\cdot}4}{t} \text{ N/mm}^2$$

Due to the vertical component W_V, the following two stresses are induced in the arm :

1. Direct shear stress,

$$\tau = \frac{W_V}{A} = \frac{20\ 000}{130\ t} = \frac{154}{t} \text{ N/mm}^2$$

2. Bending stress causing tensile stress in the upper most fibres of the arm and compressive in the lower most fibres of the arm.

Assuming that the arm extends upto the plate used for fixing the bracket to the structure. This assumption gives stronger section for the arm of the bracket.

$\therefore$ Bending moment due to W_V,

$$M_V = W_V\,(175 + 25) = 20\ 000 \times 200 = 4 \times 10^6 \text{ N-mm}$$

and bending stress, $$\sigma_{t3} = \frac{M_V}{Z} = \frac{4 \times 10^6}{2817\ t} = \frac{1420}{t} \text{ N/mm}^2$$

Net tensile stress induced in the upper most fibres of the arm of the bracket,

$$\sigma_t = \sigma_{t1} + \sigma_{t2} + \sigma_{t3} = \frac{266.5}{t} + \frac{430.4}{t} + \frac{1420}{t} = \frac{2116.9}{t} \text{ N/mm}^2 \quad ...(v)$$

We know that maximum tensile stress $[\sigma_{t(max)}]$,

$$70 = \frac{1}{2}\sigma_t + \frac{1}{2}\sqrt{(\sigma_t)^2 + 4\tau^2}$$

$$= \frac{1}{2} \times \frac{2116.9}{t} + \frac{1}{2}\sqrt{\left(\frac{2116.9}{t}\right)^2 + 4\left(\frac{154}{t}\right)^2}$$

$$= \frac{1058.45}{t} + \frac{1069.6}{t} = \frac{2128.05}{t}$$

$\therefore \quad t = 2128.05 / 70 = 30.4$ say 31 mm **Ans.**

Let us now check the shear stress induced in the arm. We know that maximum shear stress,

$$\tau_{max} = \frac{1}{2}\sqrt{(\sigma_t)^2 + 4\tau^2} = \frac{1}{2}\sqrt{\left(\frac{2116.9}{t}\right)^2 + 4\left(\frac{154}{t}\right)^2}$$

$$= \frac{1069.6}{t} = \frac{1069.6}{31} = 34.5 \text{ N/mm}^2 = 34.5 \text{ MPa}$$

Since the induced shear stress is less than the permissible stress (50 MPa), therefore the design is safe.

15. Design and make a neat dimensioned sketch of a muff coupling which is used to connect two steel shafts transmitting 40 kW at 350 r.p.m. The material for the shafts and key is plain carbon steel for which allowable shear and crushing stresses may be taken as 40 MPa and 80 MPa respectively. The material for the muff is cast iron for which the allowable shear stress may be assumed as 15 MPa.

Given: P = 40 kW = 40 × 10^3 W; N = 350 r.p.m.; τs = 40 MPa = 40 N/mm2; σ_{cs} = 80 MPa = 80 N/mm^2; τ_c = 15 MPa = 15 N/mm^2

1. *Design for shaft*

Let d = Diameter of the shaft.

We know that the torque transmitted by the shaft, key and muff,

$$T = \frac{P \times 60}{2\pi N} = \frac{40 \times 10^3 \times 60}{2\pi \times 350} = 1100 \text{ N-m}$$

$$= 1100 \times 10^3 \text{ N-mm}$$

We also know that the torque transmitted (T),

$$1100 \times 10^3 = \frac{\pi}{16} \times \tau_s \times d^3 = \frac{\pi}{16} \times 40 \times d^3 = 7.86\, d^3$$

$\therefore \quad d^3 = 1100 \times 10^3/7.86 = 140 \times 10^3$ or $d = 52$ say 55 mm **Ans.**

2. *Design for sleeve*

We know that outer diameter of the muff,

$$D = 2d + 13 \text{ mm} = 2 \times 55 + 13 = 123 \text{ say } 125 \text{ mm } \textbf{Ans.}$$

and length of the muff,

$$L = 3.5\, d = 3.5 \times 55 = 192.5 \text{ say } 195 \text{ mm } \textbf{Ans.}$$

Let us now check the induced shear stress in the muff. Let τ_c be the induced shear stress in the muff which is made of cast iron. Since the muff is considered to be a hollow shaft, therefore the torque transmitted (T),

$$1100 \times 10^3 = \frac{\pi}{16} \times \tau_c \left(\frac{D^4 - d^4}{D} \right) = \frac{\pi}{16} \times \tau_c \left[\frac{(125)^4 - (55)^4}{125} \right]$$

$$= 370 \times 10^3 \, \tau_c$$

$$\therefore \qquad \tau_c = 1100 \times 10^3 / 370 \times 10^3 = 2.97 \text{ N/mm}^2$$

Since the induced shear stress in the muff (cast iron) is less than the permissible shear stress of 15 N/mm^2, therefore the design of muff is safe.

3. *Design for key*

From Table 13.1, we find that for a shaft of 55 mm diameter,

Width of key, $w = 18$ mm **Ans.**

Since the crushing stress for the key material is twice the shearing stress, therefore a square key may be used.

$\therefore$ Thickness of key, $\quad t = w = 18$ mm **Ans.**

We know that length of key in each shaft,

$$l = L / 2 = 195 / 2 = 97.5 \text{ mm } \textbf{Ans.}$$

Let us now check the induced shear and crushing stresses in the key. First of all, let us consider shearing of the key. We know that torque transmitted (T),

$$1100 \times 10^3 = l \times w \times \tau_s \times \frac{d}{2} = 97.5 \times 18 \times \tau_s \times \frac{55}{2} = 48.2 \times 10^3 \, \tau_s$$

$$\therefore \qquad \tau_s = 1100 \times 10^3 / 48.2 \times 10^3 = 22.8 \text{ N/mm}^2$$

Now considering crushing of the key. We know that torque transmitted (T),

$$1100 \times 10^3 = l \times \frac{t}{2} \times \sigma_{cs} \times \frac{d}{2} = 97.5 \times \frac{18}{2} \times \sigma_{cs} \times \frac{55}{2} = 24.1 \times 10^3 \, \sigma_{cs}$$

$$\therefore \qquad \sigma_{cs} = 1100 \times 10^3 / 24.1 \times 10^3 = 45.6 \text{ N/mm}^2$$

Since the induced shear and crushing stresses are less than the permissible stresses, therefore the design of key is safe.

16. A single plate clutch, effective on both sides, is required to transmit 25 kW at 3000 r.p.m. Determine the outer and inner diameters of frictional surface if the coefficient of friction is 0.255, ratio of diameters is 1.25 and the maximum pressure is not to exceed 0.1 N/mm2. Also, determine the axial thrust to be provided by springs. Assume the theory of uniform wear.

Solution. Given : $n = 2$; $P = 25$ kW $= 25 \times 10^3$ W ; $N = 3000$ r.p.m. ; $\mu = 0.255$; $d_1 / d_2 = 1.25$ or $r_1 / r_2 = 1.25$; $p_{max} = 0.1$ N/mm^2

Outer and inner diameters of frictional surface

Let d_1 and d_2 = Outer and inner diameters (in mm) of frictional surface, and

r_1 and r_2 = Corresponding radii (in mm) of frictional surface.

We know that the torque transmitted by the clutch,

$$T = \frac{P \times 60}{2\,\pi\,N} = \frac{25 \times 10^3 \times 60}{2\,\pi \times 3000} = 79.6 \text{ N-m} = 79\,600 \text{ N-mm}$$

For uniform wear conditions, $p.r = C$ (a constant). Since the intensity of pressure is maximum at the inner radius (r_2), therefore.

$$p_{max} \times r_2 = C$$

or $$C = 0.1\, r_2 \text{ N/mm}$$

and normal or axial load acting on the friction surface,

$$W = 2\pi\, C\,(r_1 - r_2) = 2\pi \times 0.1\, r_2\,(1.25\, r_2 - r_2)$$
$$= 0.157\,(r_2)^2 \qquad \ldots (\because\ r_1 / r_2 = 1.25)$$

We know that mean radius of the frictional surface (for uniform wear),

$$R = \frac{r_1 + r_2}{2} = \frac{1.25\, r_2 + r_2}{2} = 1.125\, r_2$$

and the torque transmitted (T),

$$79\,600 = n.\mu.W.R = 2 \times 0.255 \times 0.157\,(r_2)^2\, 1.125\, r_2 = 0.09\,(r_2)^3$$

$\therefore$ $$(r_2)^3 = 79.6 \times 10^3 / 0.09 = 884 \times 10^3 \text{ or } r_2 = 96 \text{ mm}$$

and $$r_1 = 1.25\, r_2 = 1.25 \times 96 = 120 \text{ mm}$$

$\therefore$ Outer diameter of frictional surface,

$$d_1 = 2r_1 = 2 \times 120 = 240 \text{ mm}$$ **Ans.**

and inner diameter of frictional surface,

$$d_2 = 2r_2 = 2 \times 96 = 192 \text{ mm}$$ **Ans.**

Axial thrust to be provided by springs

We know that axial thrust to be provided by springs,

$$W = 2\pi\, C\,(r_1 - r_2) = 2\pi \times 0.1\, r_2\,(1.25\, r_2 - r_2)$$
$$= 0.157\,(r_2)^2 = 0.157\,(96)^2 = 1447 \text{ N}$$ **Ans.**

17. A helical spring is made from a wire of 6 mm diameter and has outside diameter of 75 mm. If the permissible shear stress is 350 MPa and modulus of rigidity 84 kN/mm^2, find the axial load which the spring can carry and the deflection per active turn.

Solution. Given : $d = 6$ mm ; $D_o = 75$ mm ; $\tau = 350$ MPa $= 350$ N/mm^2 ; $G = 84$ kN/mm^2 $= 84 \times 10^3$ N/mm^2

We know that mean diameter of the spring,

$$D = D_o - d = 75 - 6 = 69 \text{ mm}$$

$\therefore$ Spring index, $$C = \frac{D}{d} = \frac{69}{6} = 11.5$$

Let W = Axial load, and

δ / n = Deflection per active turn.

1. ***Neglecting the effect of curvature***

We know that the shear stress factor,

$$K_S = 1 + \frac{1}{2C} = 1 + \frac{1}{2 \times 11.5} = 1.043$$

and maximum shear stress induced in the wire (τ),

$$350 = K_S \times \frac{8\,W.D}{\pi\,d^3} = 1.043 \times \frac{8\,W \times 69}{\pi \times 6^3} = 0.848\,W$$

$\therefore$ $W = 350 / 0.848 = 412.7$ N **Ans.**

We know that deflection of the spring,

$$\delta = \frac{8\,W.D^3.n}{G.d^4}$$

$\therefore$ Deflection per active turn,

$$\frac{\delta}{n} = \frac{8\,W.D^3}{G.d^4} = \frac{8 \times 412.7\,(69)^3}{84 \times 10^3 \times 6^4} = 9.96 \text{ mm}$$ **Ans.**

2. ***Considering the effect of curvature***

We know that Wahl's stress factor,

$$K = \frac{4C-1}{4C-4} + \frac{0.615}{C} = \frac{4 \times 11.5 - 1}{4 \times 11.5 - 4} + \frac{0.615}{11.5} = 1.123$$

We also know that the maximum shear stress induced in the wire (τ),

$$350 = K \times \frac{8W.C}{\pi\,d^2} = 1.123 \times \frac{8 \times W \times 11.5}{\pi \times 6^2} = 0.913\,W$$

$\therefore$ $W = 350 / 0.913 = 383.4$ N **Ans.**

and deflection of the spring,

$$\delta = \frac{8\,W.D^3.n}{G.d^4}$$

$\therefore$ Deflection per active turn,

$$\frac{\delta}{n} = \frac{8\,W.D^3}{G.d^4} = \frac{8 \times 383.4\,(69)^3}{84 \times 10^3 \times 6^4} = 9.26 \text{ mm}$$ **Ans.**

18. **Design a knuckle joint to transmit 150 kN. The design stresses may be taken as 75 MPa in tension, 60 MPa in shear and 150 MPa in compression.**

Given: P = 150 kN = 150×10^3 N; σ_t = 75 MPa = 75 N/mm^2; τ = 60 MPa = 60 N/mm2; σ_c = 150 MPa = 150 N/mm^2

The joint is designed by considering the various methods of failure as discussed below:

1. *Failure of the solid rod in tension*

Let d = Diameter of the rod.

We know that the load transmitted (P),

$$150 \times 10^3 = \frac{\pi}{4} \times d^2 \times \sigma_t = \frac{\pi}{4} \times d^2 \times 75 = 59\, d^2$$

$$\therefore \quad d^2 = 150 \times 10^3 / 59 = 2540 \quad \text{or} \quad d = 50.4 \text{ say } 52 \text{ mm} \quad \textbf{Ans.}$$

Now the various dimensions are fixed as follows :

Diameter of knuckle pin,

$$d_1 = d = 52 \text{ mm}$$

Outer diameter of eye, $d_2 = 2\, d = 2 \times 52 = 104$ mm

Diameter of knuckle pin head and collar,

$$d_3 = 1.5\, d = 1.5 \times 52 = 78 \text{ mm}$$

Thickness of single eye or rod end,

$$t = 1.25\, d = 1.25 \times 52 = 65 \text{ mm}$$

Thickness of fork, $t_1 = 0.75\, d = 0.75 \times 52 = 39$ say 40 mm

Thickness of pin head, $t_2 = 0.5\, d = 0.5 \times 52 = 26$ mm

2. *Failure of the knuckle pin in shear*

Since the knuckle pin is in double shear, therefore load (P),

$$150 \times 10^3 = 2 \times \frac{\pi}{4} \times (d_1)^2\, \tau = 2 \times \frac{\pi}{4} \times (52)^2\, \tau = 4248\, \tau$$

$$\therefore \quad \tau = 150 \times 10^3 / 4248 = 35.3 \text{ N/mm}^2 = 35.3 \text{ MPa}$$

3. *Failure of the single eye or rod end in tension*

The single eye or rod end may fail in tension due to the load. We know that load (P),

$$150 \times 10^3 = (d_2 - d_1)\, t \times \sigma_t = (104 - 52)\, 65 \times \sigma_t = 3380\, \sigma_t$$

$$\therefore \quad \sigma_t = 150 \times 10^3 / 3380 = 44.4 \text{ N/mm}^2 = 44.4 \text{ MPa}$$

4. *Failure of the single eye or rod end in shearing*

The single eye or rod end may fail in shearing due to the load. We know that load (P),

$$150 \times 10^3 = (d_2 - d_1)\, t \times \tau = (104 - 52)\, 65 \times \tau = 3380\, \tau$$

$$\therefore \quad \tau = 150 \times 10^3 / 3380 = 44.4 \text{ N/mm}^2 = 44.4 \text{ MPa}$$

5. *Failure of the single eye or rod end in crushing*

The single eye or rod end may fail in crushing due to the load. We know that load (P),

$$150 \times 10^3 = d_1 \times t \times \sigma_c = 52 \times 65 \times \sigma_c = 3380\, \sigma_c$$

$$\therefore \quad \sigma_c = 150 \times 10^3 / 3380 = 44.4 \text{ N/mm}^2 = 44.4 \text{ MPa}$$

6. *Failure of the forked end in tension*

The forked end may fail in tension due to the load. We know that load (P),

$$150 \times 10^3 = (d_2 - d_1)\, 2\, t_1 \times \sigma_t = (104 - 52)\, 2 \times 40 \times \sigma_t = 4160\, \sigma_t$$

$$\therefore \quad \sigma_t = 150 \times 10^3 / 4160 = 36 \text{ N/mm}^2 = 36 \text{ MPa}$$

7. *Failure of the forked end in shear*

The forked end may fail in shearing due to the load. We know that load (P),

$$150 \times 10^3 = (d_2 - d_1)\, 2\, t_1 \times \tau = (104 - 52)\, 2 \times 40 \times \tau = 4160\, \tau$$

$$\therefore \quad \tau = 150 \times 10^3 / 4160 = 36 \text{ N/mm}^2 = 36 \text{ MPa}$$

8. *Failure of the forked end in crushing*

The forked end may fail in crushing due to the load. We know that load (P),

$$150 \times 10^3 = d_1 \times 2\, t_1 \times \sigma_c = 52 \times 2 \times 40 \times \sigma_c = 4160\, \sigma_c$$

$$\therefore \quad \sigma_c = 150 \times 10^3 / 4180 = 36 \text{ N/mm}^2 = 36 \text{ MPa}$$

From above, we see that the induced stresses are less than the given design stresses, therefore the joint is safe.

19. A shaft supported at the ends in ball bearings carries a straight tooth spur gear at its mid span and is to transmit 7.5 kW at 3000 r.p.m. The pitch circle diameter of the gear is 150 mm. The distances between the centre line of bearings and gear are 100 mm each. If the shaft is made of steel and the allowable shear stress is 45 MPa, determine the diameter of the shaft. Show in a sketch how the gear will be mounted on the shaft; also indicate the ends whe re the bearings will be mounted? The pressure angle of the gear may be taken as 20°.

Given: P = 7.5 kW = 7500 W; N = 3000 r.p.m.; D = 150 mm = 0.15 m; L = 200 mm = 0.2 m; τ = 45 MPa = 45 N/mm^2; α = 20°

We know that torque transmitted by the shaft,

We know that torque transmitted by the shaft,

$$T = \frac{P \times 60}{2\pi N} = \frac{7500 \times 60}{2\pi \times 300} = 23.8 \text{ N-m}$$

$\therefore$ Tangential force on the gear,

$$F_t = \frac{2T}{D} = \frac{2 \times 238.7}{0.15} = 318.27 \text{ N}$$

and the normal load acting on the tooth of the gear,

$$W = \frac{F_t}{\cos\alpha} = \frac{318.27}{\cos 20^\circ} = \frac{318.27}{0.9397} = 338.7 \text{ N}$$

Since the gear is mounted at the middle of the shaft, therefore maximum bending moment at the centre of the gear,

$$M = \frac{W.L}{4} = \frac{338.7 \times 0.2}{4} = 16.94 \text{ N-m}$$

Let d = Diameter of the shaft.

We know that equivalent twisting moment,

$$T_e = \sqrt{M^2 + T^2} = \sqrt{(16.94)^2 + (23.87)^2} = 29.27 \text{ N-m}$$
$$= 29.27 \times 10^3 \text{ N-mm}$$

We also know that equivalent twisting moment (T_e),

$$29.27 \times 10^3 = \frac{\pi}{16} \times \tau \times d^3 = \frac{\pi}{16} \times 45 \times d^3 = 8.84 \ d^3$$

$\therefore$ $d^3 = 29.27 \times 10^3 / 8.84 = 3.3 \times 10^3$ or $d = 14.8$ say 15 mm **Ans.**

20. Determine the diameter of a circular rod made of ductile material with a fatigue strength (complete stress reversal), σ_e = 265 MPa and a tensile yield strength of 350 MPa. The member is subjected to a varying axial load from W_{min} = – 300 × 10^3 N to W_{max} = 700 × 10^3 N and has a stress concentration factor = 1.8. Use factor of safety as 2.0.

Given: σ_e = 265 MPa = 265 N/mm^2 ; σ_y = 350 MPa = 350 N/mm^2; W_{min} = – 300 × 10^3 N; W_{max} = 700 × 10^3 N ; K_f = 1.8 ; F.S. = 2

Let d = Diameter of the circular rod in mm.

$\therefore$ Area, $A = \frac{\pi}{4} \times d^2 = 0.7854 \ d^2 \text{ mm}^2$

We know that the mean or average load,

$$W_m = \frac{W_{max} + W_{min}}{2} = \frac{700 \times 10^3 + (-300 \times 10^3)}{2} = 200 \times 10^3 \text{ N}$$

$\therefore$ Mean stress, $\sigma_m = \frac{W_m}{A} = \frac{200 \times 10^3}{0.7854 \ d^2} = \frac{254.6 \times 10^3}{d^2} \text{ N/mm}^2$

Variable load, $W_v = \frac{W_{max} - W_{min}}{2} = \frac{700 \times 10^3 - (-300 \times 10^3)}{2} = 500 \times 10^3 \text{ N}$

$\therefore$ Variable stress, $\sigma_v = \frac{W_v}{A} = \frac{500 \times 10^3}{0.7854 \ d^2} = \frac{636.5 \times 10^3}{d^2} \text{ N/mm}^2$

We know that according to Soderberg's formula,

$$\frac{1}{F.S.} = \frac{\sigma_m}{\sigma_y} + \frac{\sigma_v \times K_f}{\sigma_e}$$

$$\frac{1}{2} = \frac{254.6 \times 10^3}{d^2 \times 350} + \frac{636.5 \times 10^3 \times 1.8}{d^2 \times 265} = \frac{727}{d^2} + \frac{4323}{d^2} = \frac{5050}{d^2}$$

$\therefore$ $d^2 = 5050 \times 2 = 10\,100$ or $d = 100.5$ mm **Ans.**

5615121

B.TECH. DEGREE EXAMINATION APRIL/MAY 2017

Fifth semester – Mechanical Engineering

DESIGN OF MACHINE ELEMENTS

1. What is adoptive design?

In most cases, the designer's work is concerned with adaptation of existing designs. This type of design needs no special knowledge or skill and can be attempted by designers of ordinary technical training. The designer only makes minor alternation or modification in the existing designs of the product.

2. State any four important factors that influence manufacturing.

- Availability of raw materials
- Availability of Labour
- Transport Facilities
- Power
- Finance
- Natural and Climatic Considerations

3. Why V threads are used in power screw?

Power screws are classified by the geometry of their thread. V-threads are less suitable for lead screws than others such as ACME because they have more friction between the threads. Their threads are designed to induce this friction to keep the fastener from loosening.

4. How are coarse series threads designated?

Coarse thread is designated by the capital letter M followed by the major diameter of the thread in mm. For example M10

5. What are the main functions of key?

The main function key is to connect two machine parts for preventing relative motion of rotation with respect to each other.

6. Give atleast three practical applications of couplings.

Couplings are used in various industries like textiles, beverages, dairies, semiconductor production, aggregate production, food processing industry etc.

7. Distinguish between a closely coiled and open coiled helical spring.

Closely coiled	**Open coiled**
Gap between successive turns is small	Gap between successive turns is large
Helix angle is small and is generally, less than 10^0	Helix angle is small and is generally, less than 10^0
Stresses upon the material are almost of pure torsion	Bending stresses are not negligible compared o the Torsional stresses.
Can take up tensile load only	Can take up compressive loads also
Frictional effect is more	Frictional effect is less

8. Where are knuckle and cotter joints used?

Knuckle joint are used in the link of a cycle chain, tie rod joint for roof truss, valve rod joint with eccentric rod, pump rod joint, tension link in bridge structure and lever and rod connections of various types.

Cotter joints are **used in** connecting a piston rod to the crosshead of a reciprocating steam engine, a piston rod and its extension as a tail or pump rod, strap end of connecting rod etc.

9. What are various types of stresses induced in the shafts?

- ✓ Shear stresses due to transmission of torque.
- ✓ Bending stresses.
- ✓ Stresses due to combined torsional and bending loads.

10. What do you mean by fatigue failure?

Fatigue failure is defined as the tendency of a material to fracture by means of progressive brittle cracking under repeated alternating or cyclic stresses of an intensity considerably below the normal strength.

Part B

11. The stresses induced at a critical point in a machine component made of steel 45C8 with a yield strength of 380N/mm^2 is as follows: σ_x= 100 N/mm^2 σ_y= 40 N/mm^2 τ_{xy}= 100 N/mm^{2}1. Maximum principal stress theory; 2. Maximum shear stress theory; 3. Maximum distortion energy theory.

Given : $\sigma_x = 100\ N/mm^2$ $\sigma_y = 40\ N/mm^2$ $\tau_{xy} = 80\ N/mm^2$

a) Max normal stress theory $\sigma = \frac{\sigma_x + \sigma_y}{2} + \frac{1}{2}\left[\sqrt{(\sigma_x - \sigma_y)^2 + 4\tau_{xy}^2}\right]$

$$\sigma_t = \frac{100+40}{2} + \frac{1}{2}\left[\sqrt{(100-40)^2 + 4 \times 80^2}\right]$$

$$\frac{\sigma_{yt}}{F.O.S} = \sigma_t \Rightarrow \frac{380}{F.O.S} = 155 \quad F.O.S = 2.4 \simeq 2.5$$

b) Max. Shear stress theory

$$\tau_{max} = \frac{\sigma_x - \sigma_y}{2} = \frac{60}{2} = 30$$

$$\tau_{max} = \frac{\sigma_{yt}}{2 \times F.O.S} \Rightarrow 30 = \frac{380}{2 \times F.O.S}$$

$$F.O.S = 6$$

c) Distortion energy theory

$$\sigma_x^2 + \sigma_y^2 - \sigma_x\sigma_y = \sigma_y^2$$

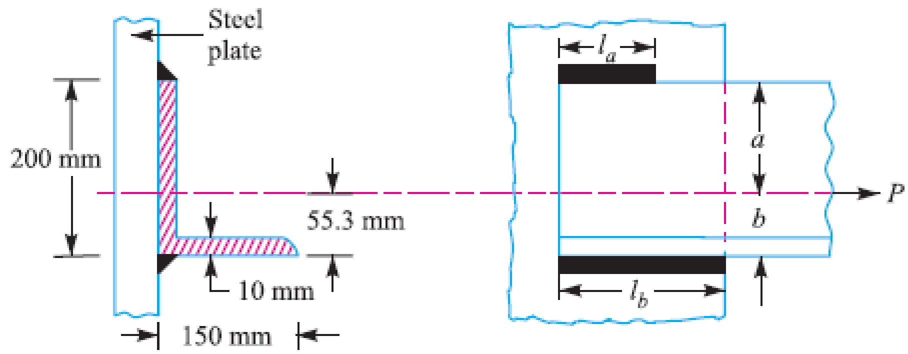
$$100^2 + 40^2 - 100 \times 40 = \frac{380^2}{F.OS \times 2}$$

$$\cancel{7600 = \frac{144400}{FOS}}$$

$$7600 = \frac{72200}{F.OS}$$

$$F.O.S = 9$$

12. **A 200 × 150 × 10 mm angle is to be welded to a steel plate by fillet welds as shown in Fig. If the angle is subjected to a static load of 200 kN, find the length of weld at the top and bottom. The allowable shear stress for static loading may be taken as 75 MPa.**



Solution. Given : $a + b = 200$ mm ; $P = 200$ kN $= 200 \times 10^3$ N ; $\tau = 75$ MPa $= 75$ N/mm^2

Let l_a = Length of weld at the top,

l_b = Length of weld at the bottom, and

l = Total length of the weld $= l_a + l_b$

Since the thickness of the angle is 10 mm, therefore size of weld,

$$s = 10 \text{ mm}$$

We know that for a single parallel fillet weld, the maximum load (P),

$$200 \times 10^3 = 0.707\, s \times l \times \tau = 0.707 \times 10 \times l \times 75 = 530.25\, l$$

$$\therefore \quad l = 200 \times 10^3 / 530.25 = 377 \text{ mm}$$

or $$l_a + l_b = 377 \text{ mm}$$

Now let us find out the position of the centroidal axis.

Let b = Distance of centroidal axis from the bottom of the angle.

$$\therefore \quad b = \frac{(200 - 10)\, 10 \times 95 + 150 \times 10 \times 5}{190 \times 10 + 150 \times 10} = 55.3 \text{ mm}$$

and $$a = 200 - 55.3 = 144.7 \text{ mm}$$

We know that $$l_a = \frac{l \times b}{a + b} = \frac{377 \times 55.3}{200} = 104.2 \text{ mm} \quad \textbf{Ans.}$$

and $$l_b = l - l_a = 377 - 104.2 = 272.8 \text{ mm} \quad \textbf{Ans.}$$

13. **A 50 kN capacity screw jack consists of square threaded steel screw meshing with a bronze nut. The nominal diameter is 60 mm and the pitch is 9 mm. the permissible bearing pressure at the threads is 10 N/mm^2.Calculate**

   a) The length of the nut

   b) The shear stress in the nut.

Given : $W = 50 \times 10^3$ N ; $d = 60$ mm ; $S = 9$ mm ; $p = 10$ N/mm^2

W.K.T $$p = \frac{W}{\pi \times d \times t \times n}$$

$n \to$ no. of screw Eng. the screw

$$10 = \frac{50 \times 10^3}{\pi \times 60 \times 4.5 \times n}$$

$$t = \frac{S}{2} = \frac{9}{2}$$

$$n = 5.8 \simeq 6$$

a) Length of the nut

$$h = n \times \text{pitch}(s)$$

$$h = 54 \text{ mm}.$$

b) Shear stress in the nut

$$\tau_{max} = \frac{1}{2}\sqrt{\sigma_c^2 + 4\tau^2}$$

$$d_c = d - S/2$$
$$= 60 -$$

$$\sigma = \frac{W}{\frac{\pi}{4}(d_c)^2} = \frac{50 \times 10^3}{\pi/4 \times 55^2}$$

$$\sigma = 20.6 \text{ N/mm}^2$$

$$T = \pi/16 \, \tau d^3$$

$$T = W \times \frac{d}{2}$$
$$T = 1500 \times 10^3 \text{ N.m}$$

$$1500 \times 10^3 = \frac{\pi}{16} \tau \times 60^3$$

$$\tau = 35.3 \text{ N/mm}^2$$

$$\tau_{max} = \frac{1}{2}\sqrt{20^2 + 4(35.3)^2}$$

$$\tau_{max} = 36.68 \text{ N/mm}^2$$

14. A steam engine cylinder has an effective diameter of 350 mm and the maximum steam pressure acting on the cylinder cover is 1.25 N/mm². Calculate the number and size of studs required to fix the cylinder cover, assuming the permissible stress in the studs as 33 MPa.

Solution. Given: $D = 350$ mm ; $p = 1.25$ N/mm^2 ; $\sigma_t = 33$ MPa $= 33$ N/mm^2

Let d = Nominal diameter of studs,

d_c = Core diameter of studs, and

n = Number of studs.

We know that the upward force acting on the cylinder cover,

$$P = \frac{\pi}{4} \times D^2 \times p = \frac{\pi}{4}(350)^2\ 1.25 = 120\ 265 \text{ N} \quad ...(i)$$

Assume that the studs of nominal diameter 24 mm are used. From Table 11.1 (coarse series), we find that the corresponding core diameter (d_c) of the stud is 20.32 mm.

∴ Resisting force offered by n number of studs,

$$P = \frac{\pi}{4} \times (d_c)^2\ \sigma_t \times n = \frac{\pi}{4}(20.32)^2\ 33 \times n = 10\ 700\ n\text{N} \quad ...(ii)$$

From equations *(i)* and *(ii)*, we get

$$n = 120\ 265 / 10\ 700 = 11.24 \text{ say } 12 \textbf{ Ans.}$$

Taking the diameter of the stud hole (d_1) as 25 mm, we have pitch circle diameter of the studs,

$$D_p = D + 2t + 3d_1 = 350 + 2 \times 10 + 3 \times 25 = 445 \text{ mm}$$

...(Assuming t = 10 mm)

$\therefore$ *Circumferential pitch of the studs

$$= \frac{\pi \times D_p}{n} = \frac{\pi \times 445}{12} = 116.5 \text{ mm}$$

We know that for a leak-proof joint, the circumferential pitch of the studs should be between $20\sqrt{d_1}$ to $30\sqrt{d_1}$, where d_1 is the diameter of stud hole in mm.

$\therefore$ Minimum circumferential pitch of the studs

$$= 20\sqrt{d_1} = 20\sqrt{25} = 100 \text{ mm}$$

and maximum circumferential pitch of the studs

$$= 30\sqrt{d_1} = 30\sqrt{25} = 150 \text{ mm}$$

Since the circumferential pitch of the studs obtained above lies within 100 mm to 150 mm, therefore the size of the stud chosen is satisfactory.

$\therefore$ Size of the stud = M 24 **Ans.**

15. Design a cast iron protective type flange coupling to transmit 15 kW at 900r.p.m. from an electric motor to a compressor. The service factor may be assumed as 1.35. The following permissible stresses may be used: Shear stress for shaft, bolt and key material = 40 Mpa, Crushing stress for bolt and key = 80 Mpa, Shear stress for cast iron = 8 Mpa, Draw a neat sketch of the coupling.

Solution. Given : P = 15 kW = 15×10^3 W ; N = 900 r.p.m. ; Service factor = 1.35 ; $\tau_s = \tau_b = \tau_k$ = 40 MPa = 40 N/mm^2; $\sigma_{cb} = \sigma_{ck}$ = 80 MPa = 80 N/mm^2 ; τ_c = 8 MPa = 8 N/mm^2

The protective type flange coupling is designed as discussed below :

1. Design for hub

First of all, let us find the diameter of the shaft (d). We know that the torque transmitted by the shaft,

$$T = \frac{P \times 60}{2\pi N} = \frac{15 \times 10^3 \times 60}{2\pi \times 900} = 159.13 \text{ N-m}$$

Since the service factor is 1.35, therefore the maximum torque transmitted by the shaft,

$$T_{max} = 1.35 \times 159.13 = 215 \text{ N-m} = 215 \times 10^3 \text{ N-mm}$$

We know that the torque transmitted by the shaft (T),

$$215 \times 10^3 = \frac{\pi}{16} \times \tau_s \times d^3 = \frac{\pi}{16} \times 40 \times d^3 = 7.86\, d^3$$

$\therefore$ $d^3 = 215 \times 10^3 / 7.86 = 27.4 \times 10^3$ or d = 30.1 say 35 mm **Ans.**

We know that outer diameter of the hub,

$$D = 2d = 2 \times 35 = 70 \text{ mm } \textbf{Ans.}$$

and length of hub, $L = 1.5\, d = 1.5 \times 35 = 52.5$ mm **Ans.**

$$215 \times 10^3 = \frac{\pi}{16} \times \tau_c \left[\frac{D^4 - d^4}{D}\right] = \frac{\pi}{16} \times \tau_c \left[\frac{(70)^4 - (35)^4}{70}\right] = 63\,147\, \tau_c$$

$\therefore$ $\tau_c = 215 \times 10^3 / 63\,147 = 3.4$ N/mm^2 = 3.4 MPa

Since the induced shear stress for the hub material (*i.e.* cast iron) is less than the permissible value of 8 MPa, therefore the design of hub is safe.

2. *Design for key*

Since the crushing stress for the key material is twice its shear stress (*i.e.* $\sigma_{ck} = 2\tau_k$), therefore a square key may be used. From Table 13.1, we find that for a shaft of 35 mm diameter,

Width of key, $w = 12$ mm **Ans.**

and thickness of key, $t = w = 12$ mm **Ans.**

The length of key (l) is taken equal to the length of hub.

$\therefore \quad l = L = 52.5$ mm **Ans.**

Let us now check the induced stresses in the key by considering it in shearing and crushing.

Considering the key in shearing. We know that the maximum torque transmitted (T_{max}),

$$215 \times 10^3 = l \times w \times \tau_k \times \frac{d}{2} = 52.5 \times 12 \times \tau_k \times \frac{35}{2} = 11\ 025\ \tau_k$$

$$\therefore \quad \tau_k = 215 \times 10^3 / 11\ 025 = 19.5 \text{ N/mm}^2 = 19.5 \text{ MPa}$$

Considering the key in crushing. We know that the maximum torque transmitted (T_{max}),

$$215 \times 10^3 = l \times \frac{t}{2} \times \sigma_{ck} \times \frac{d}{2} = 52.5 \times \frac{12}{2} \times \sigma_{ck} \times \frac{35}{2} = 5512.5\ \sigma_{ck}$$

$$\therefore \quad \sigma_{ck} = 215 \times 10^3 / 5512.5 = 39 \text{ N/mm}^2 = 39 \text{ MPa}$$

Since the induced shear and crushing stresses in the key are less than the permissible stresses, therefore the design for key is safe.

3. *Design for flange*

The thickness of flange (t_f) is taken as 0.5 d.

$\therefore \quad t_f = 0.5\ d = 0.5 \times 35 = 17.5$ mm **Ans.**

Let us now check the induced shearing stress in the flange by considering the flange at the junction of the hub in shear.

We know that the maximum torque transmitted (T_{max}),

$$215 \times 10^3 = \frac{\pi\ D^2}{2} \times \tau_c \times t_f = \frac{\pi (70)^2}{2} \times \tau_c \times 17.5 = 134\ 713\ \tau_c$$

$$\therefore \quad \tau_c = 215 \times 10^3 / 134\ 713 = 1.6 \text{ N/mm}^2 = 1.6 \text{ MPa}$$

Since the induced shear stress in the flange is less than 8 MPa, therefore the design of flange is safe.

4. *Design for bolts*

Let $\quad d_1$ = Nominal diameter of bolts.

Since the diameter of the shaft is 35 mm, therefore let us take the number of bolts,

$$n = 3$$

and pitch circle diameter of bolts,

$$D_1 = 3d = 3 \times 35 = 105 \text{ mm}$$

The bolts are subjected to shear stress due to the torque transmitted. We know that the maximum torque transmitted (T_{max}),

$$215 \times 10^3 = \frac{\pi}{4} (d_1)^2\ \tau_b \times n \times \frac{D_1}{2} = \frac{\pi}{4} (d_1)^2\ 40 \times 3 \times \frac{105}{2} = 4950\ (d_1)^2$$

$$\therefore \quad (d_1)^2 = 215 \times 10^3 / 4950 = 43.43 \text{ or } d_1 = 6.6 \text{ mm}$$

Assuming coarse threads, the nearest standard size of bolt is M 8. **Ans.**

Other proportions of the flange are taken as follows :

Outer diameter of the flange,

$$D_2 = 4\,d = 4 \times 35 = 140 \text{ mm} \quad \textbf{Ans.}$$

Thickness of the protective circumferential flange,

$$t_p = 0.25\,d = 0.25 \times 35 = 8.75 \text{ say } 10 \text{ mm} \quad \textbf{Ans.}$$

16. A solid cast iron disk 1m in diameter and 0.3 m thick is used as a flywheel. It is rotating at 350 rpm. It is brought to rest in 1.5 seconds by means of a brake.calaculate

a) The energy absorbed by the brake and

b) The torque capacity of the brake.

Given ; $D = 1\text{ m}$, $t = 0.2\text{ m}$; $n = 350$ rpm ; $t = 1.5$ S

Density of Cast iron $= 7200 \text{ kg/m}^3$

Radius of Gyration $k = d/\sqrt{8}$

$$\text{Mass} = \frac{\pi}{4} 1^2 \times 0.2^2 \times 7200$$

$$= 1130.97 \text{ kg}$$

$$K^2 = \frac{D^2}{8} = \frac{1}{8} \text{m}^2$$

$$\omega = \frac{2\pi n}{60} = \frac{2 \times \pi \times 350}{60} = 36.65 \text{ rad/sec}$$

a) Energy absorbed by brake

$$E = \frac{1}{2} m k^2 \omega^2$$

$$= \frac{1}{2}\, 1130.97 \times \frac{1}{8} \times 36.65^2$$

$$= 94946.52 \text{ J}$$

b) Torque Capacity of brake

$$\theta = \left(\frac{\omega}{2}\right) \times t$$

$$\theta = \frac{36.65}{2} \times 1.5$$

$$\theta = 27.49 \text{ rad}$$

$$T = \frac{E}{\theta} = \frac{94946.52}{27.49} = 3453.86 \text{ N.m}$$

17. Design a spring for a balance to measure 0 to 1000 N over a scale of length 80 mm. The spring is to be enclosed in a casing of 25 mm diameter. The approximate number of turns is 30. The modulus of rigidity is 85 kN/mm^2. Also calculate the maximum shear stress induced.

Solution. Given : $W = 1000$ N ; $\delta = 80$ mm ; $n = 30$; $G = 85 \text{ kN/mm}^2 = 85 \times 10^3 \text{ N/mm}^2$

Design of spring

Let D = Mean diameter of the spring coil,

d = Diameter of the spring wire, and

C = Spring index = D/d.

Since the spring is to be enclosed in a casing of 25 mm diameter, therefore the outer diameter of the spring coil ($D_o = D + d$) should be less than 25 mm.

We know that deflection of the spring (δ),

$$80 = \frac{8\,W.C^3.n}{G.d} = \frac{8 \times 1000 \times C^3 \times 30}{85 \times 10^3 \times d} = \frac{240\,C^3}{85\,d}$$

$$\therefore \quad \frac{C^3}{d} = \frac{80 \times 85}{240} = 28.3$$

Let us assume that d = 4 mm. Therefore

$$C^3 = 28.3\,d = 28.3 \times 4 = 113.2 \quad \text{or} \quad C = 4.84$$

and $D = C.d = 4.84 \times 4 = 19.36$ mm **Ans.**

We know that outer diameter of the spring coil,

$D_o = D + d = 19.36 + 4 = 23.36$ mm **Ans.**

Since the value of D_o = 23.36 mm is less than the casing diameter of 25 mm, therefore the assumed dimension, d = 4 mm is correct.

Maximum shear stress induced

We know that Wahl's stress factor,

$$K = \frac{4C-1}{4C-4} + \frac{0.615}{C} = \frac{4 \times 4.84 - 1}{4 \times 4.84 - 4} + \frac{0.615}{4.84} = 1.322$$

$\therefore$ Maximum shear stress induced,

$$\tau = K \times \frac{8\,W.C}{\pi\,d^2} = 1.322 \times \frac{8 \times 1000 \times 4.84}{\pi \times 4^2}$$

$$= 1018.2 \text{ N/mm}^2 = 1018.2 \text{ MPa} \textbf{ Ans.}$$

18. A semi-elliptical laminated spring has an overall length of 1 m and sustains a load of 70 kN at its centre. The spring has 3 full length leaves and 15 graduated leaves with a central band of 100 mm width. All the leaves are to be stressed to 400 MPa, when fully loaded. The ratio of the total spring depth to that of width is 2. E = 210 kN/mm2.

Solution. Given : $2L_1$ = 1 m = 1000 mm ; $2W$ = 70 kN or W = 35 kN = 35×10^3 N $n_F = 3$; $n_G = 15$; l = 100 mm ; σ = 400 MPa = 400 N/mm² ; E = 210 kN/mm² = 210×10^3 N/mm²

1. *Thickness and width of leaves*

Let t = Thickness of leaves, and

b = Width of leaves.

We know that the total number of leaves,

$$n = n_F + n_G = 3 + 15 = 18$$

Since it is given that ratio of the total spring depth ($n \times t$) and width of leaves is 2, therefore

$$\frac{n \times t}{b} = 2 \quad \text{or} \quad b = n \times t / 2 = 18 \times t / 2 = 9\,t$$

We know that the effective length of the leaves,

$$2L = 2L_1 - l = 1000 - 100 = 900 \text{ mm} \quad \text{or} \quad L = 900 / 2 = 450 \text{ mm}$$

Since all the leaves are equally stressed, therefore final stress (σ),

$$400 = \frac{6\,W.L}{n.b.t^2} = \frac{6 \times 35 \times 10^3 \times 450}{18 \times 9\,t \times t^2} = \frac{583 \times 10^3}{t^3}$$

$\therefore$ $t^3 = 583 \times 10^3 / 400 = 1458$ or $t = 11.34$ say 12 mm **Ans.**

and $b = 9\,t = 9 \times 12 = 108$ mm **Ans.**

2. Initial gap

We know that the initial gap (C) that should be provided between the full length and graduated leaves before the band load is applied, is given by

$$C = \frac{2\,W.L^3}{n.E.b.t^3} = \frac{2 \times 35 \times 10^3\ (450)^3}{18 \times 210 \times 10^3 \times 108\ (12)^3} = 9.04 \text{ mm} \quad \textbf{Ans.}$$

3. Load exerted on the band after the spring is assembled

We know that the load exerted on the band after the spring is assembled,

$$W_b = \frac{2\,n_F.n_G.W}{n(2n_G + 3n_F)} = \frac{2 \times 3 \times 15 \times 35 \times 10^3}{18\ (2 \times 15 + 3 \times 3)} = 4487 \text{ N} \quad \textbf{Ans.}$$

19. A solid circular shaft is subjected to a bending moment of 3000 N-m and a torque of 10 000 N-m. The shaft is made of 45 C 8 steel having ultimate tensile stress of 700 MPa and a ultimate shear stress of 500 MPa. Assuming a factor of safety as 6, determine the diameter of the shaft.

Solution. Given : $M = 3000$ N-m $= 3 \times 10^6$ N-mm ; $T = 10\ 000$ N-m $= 10 \times 10^6$ N-mm ; $\sigma_{tu} = 700$ MPa $= 700$ N/mm^2 ; $\tau_u = 500$ MPa $= 500$ N/mm^2

We know that the allowable tensile stress,

$$\sigma_t \text{ or } \sigma_b = \frac{\sigma_{tu}}{F.S.} = \frac{700}{6} = 116.7 \text{ N/mm}^2$$

and allowable shear stress,

$$\tau = \frac{\tau_u}{F.S.} = \frac{500}{6} = 83.3 \text{ N/mm}^2$$

Let d = Diameter of the shaft in mm.

According to maximum shear stress theory, equivalent twisting moment,

$$T_e = \sqrt{M^2 + T^2} = \sqrt{(3 \times 10^6)^2 + (10 \times 10^6)^2} = 10.44 \times 10^6 \text{ N-mm}$$

We also know that equivalent twisting moment (T_e),

$$10.44 \times 10^6 = \frac{\pi}{16} \times \tau \times d^3 = \frac{\pi}{16} \times 83.3 \times d^3 = 16.36\ d^3$$

$\therefore$ $d^3 = 10.44 \times 10^6 / 16.36 = 0.636 \times 10^6$ or $d = 86$ mm

According to maximum normal stress theory, equivalent bending moment,

$$M_e = \frac{1}{2}\left(M + \sqrt{M^2 + T^2}\right) = \frac{1}{2}(M + T_e)$$

$$= \frac{1}{2}(3 \times 10^6 + 10.44 \times 10^6) = 6.72 \times 10^6 \text{ N-mm}$$

We also know that the equivalent bending moment (M_e),

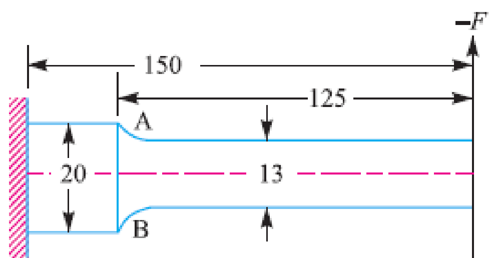
$$6.72 \times 10^6 = \frac{\pi}{32} \times \sigma_b \times d^3 = \frac{\pi}{32} \times 116.7 \times d^3 = 11.46\, d^3$$

$$\therefore \quad d^3 = 6.72 \times 10^6 / 11.46 = 0.586 \times 10^6 \text{ or } d = 83.7 \text{ mm}$$

Taking the larger of the two values, we have

d = 86 say 90 mm **Ans.**

20. A cantilever beam made of cold drawn carbon steel of circular cross-section as shown in Fig., is subjected to a load which varies from – F to 3 F. Determine the maximum load that this member can withstand for an indefinite life using a factor of safety as 2. The theoretical stress concentration factor is 1.42 and the notch sensitivity is 0.9. Assume the following values : Ultimate stress = 550 Mpa, Yield stress = 470 Mpa, Endurance limit = 275 Mpa, Size factor = 0.85, Surface finish factor=0.89



All dimensions in mm.

The beam as shown in Fig. 6.18 is subjected to a reversed bending load only. Since the point A at the change of cross section is critical, therefore we shall find the bending moment at point A.

We know that maximum bending moment at point A,

$$M_{max} = W_{max} \times 125 = 3F \times 125 = 375\, F \text{ N-mm}$$

and minimum bending moment at point A,

$$M_{min} = W_{min} \times 125 = -F \times 125 = -125\, F \text{ N-mm}$$

$\therefore$ Mean or average bending moment,

$$M_m = \frac{M_{max} + M_{min}}{2} = \frac{375\,F + (-125\,F)}{2} = 125\,F \text{ N-mm}$$

and variable bending moment,

$$M_v = \frac{M_{max} - M_{min}}{2} = \frac{375\,F - (-125\,F)}{2} = 250\,F \text{ N-mm}$$

Section modulus,

$$Z = \frac{\pi}{32} \times d^3 = \frac{\pi}{32}(13)^3 = 215.7 \text{ mm}^3 \quad \ldots(\because d = 13 \text{ mm})$$

$\therefore$ Mean bending stress,

$$\sigma_m = \frac{M_m}{Z} = \frac{125\,F}{215.7} = 0.58\,F \text{ N/mm}^2$$

and variable bending stress, $\sigma_v = \frac{M_v}{Z} = \frac{250\ F}{215.7} = 1.16\ F\ \text{N/mm}^2$

Fatigue stress concentration factor, $K_f = 1 + q\,(K_t - 1) = 1 + 0.9\,(1.42 - 1) = 1.378$

We know that according to Goodman's formula

$$\frac{1}{F.S.} = \frac{\sigma_m}{\sigma_u} + \frac{\sigma_v \times K_f}{\sigma_e \times K_{sur} \times K_{sz}}$$

$$\frac{1}{2} = \frac{0.58\ F}{550} + \frac{1.16\,F \times 1.378}{275 \times 0.89 \times 0.85}$$

$$= 0.001\ 05\ F + 0.007\ 68\ F = 0.008\ 73\ F$$

$$\therefore \qquad F = \frac{1}{2 \times 0.008\,73} = 57.3\ \text{N}$$

and according to Soderberg's formula,

$$\frac{1}{F.S.} = \frac{\sigma_m}{\sigma_y} + \frac{\sigma_v \times K_f}{\sigma_e \times K_{sur} \times K_{sz}}$$

$$\frac{1}{2} = \frac{0.58\ F}{470} + \frac{1.16\,F \times 1.378}{275 \times 0.89 \times 0.85}$$

$$= 0.001\ 23\ F + 0.007\ 68\ F = 0.008\ 91\ F$$

$$\therefore \qquad F = \frac{1}{2 \times 0.008\ 91} = 56\ \text{N}$$

Taking larger of the two values, we have $F = 57.3$ N **Ans.**